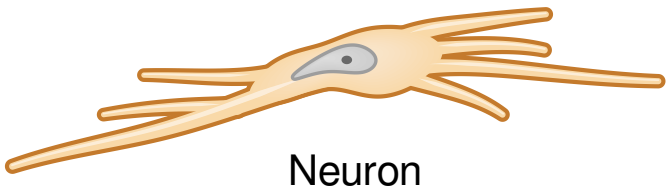
On the precision of dynamic gene regulation

Tarek Tohme

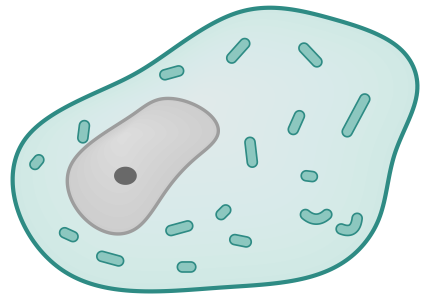
Under the supervision of

Aleksandra Walczak and Thierry Mora

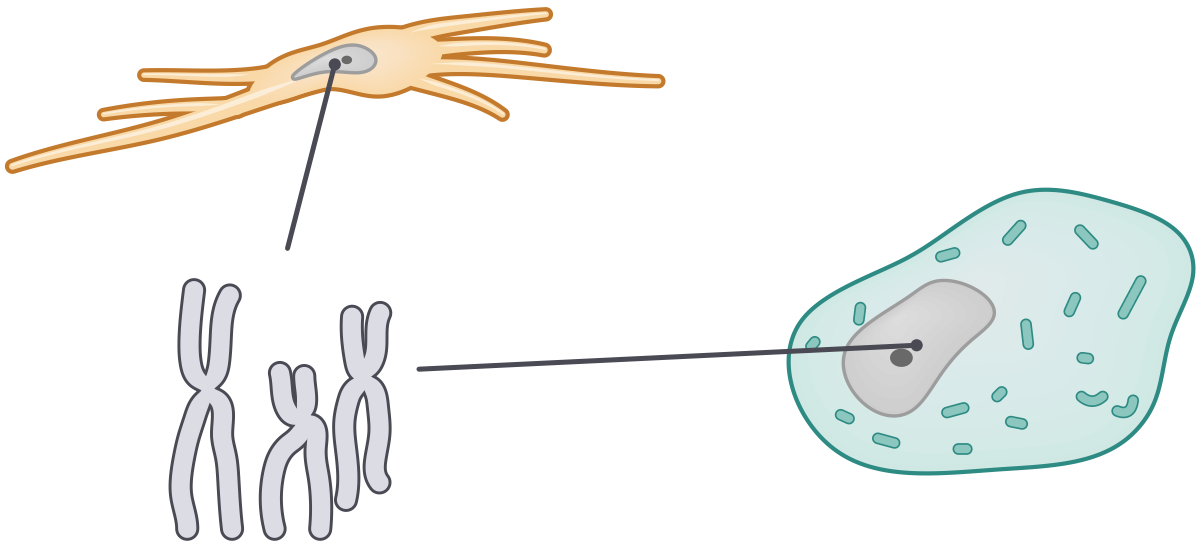
November 6, 2026



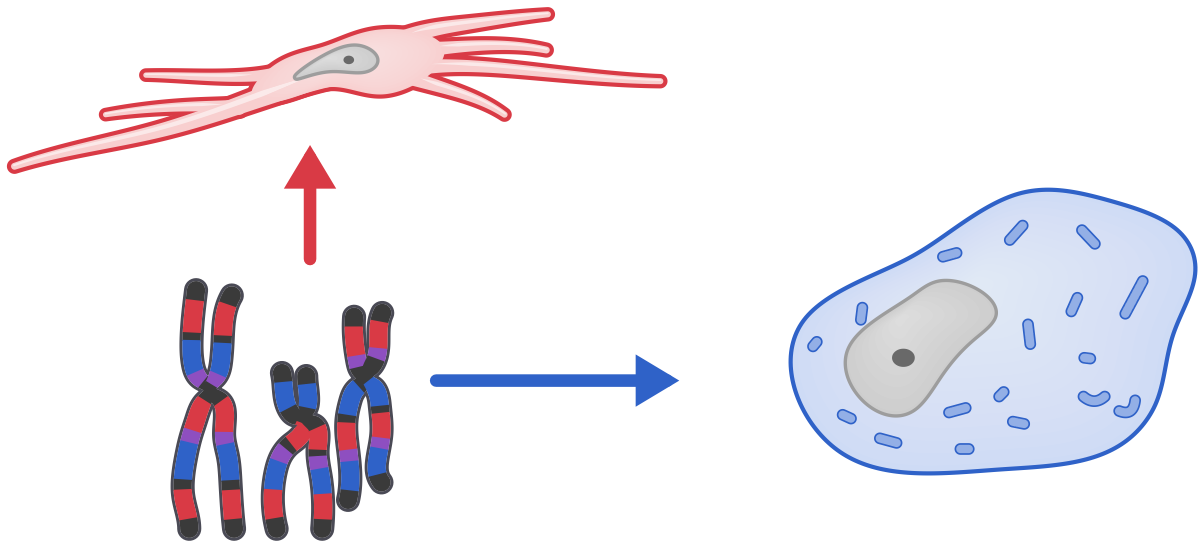
Neuron



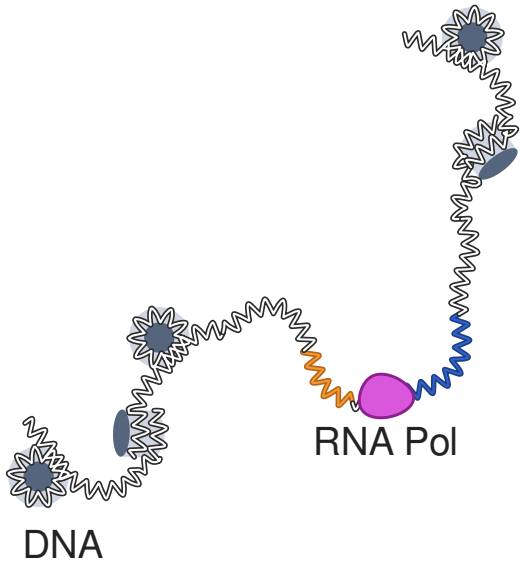
Kidney cell



Gene expression is regulated



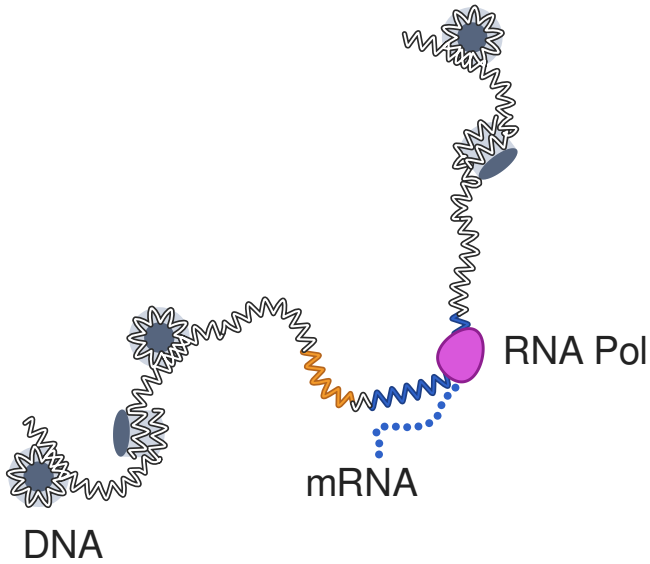
Overview of gene regulation



Transcription

DNA
Promoter
Gene
RNA Polymerase

Overview of gene regulation



Transcription

DNA
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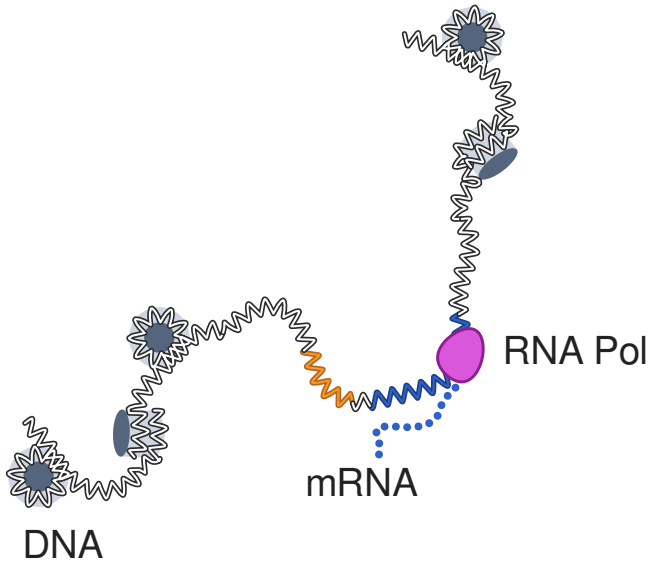
mRNA

Overview of gene regulation



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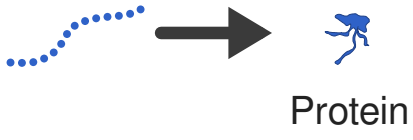
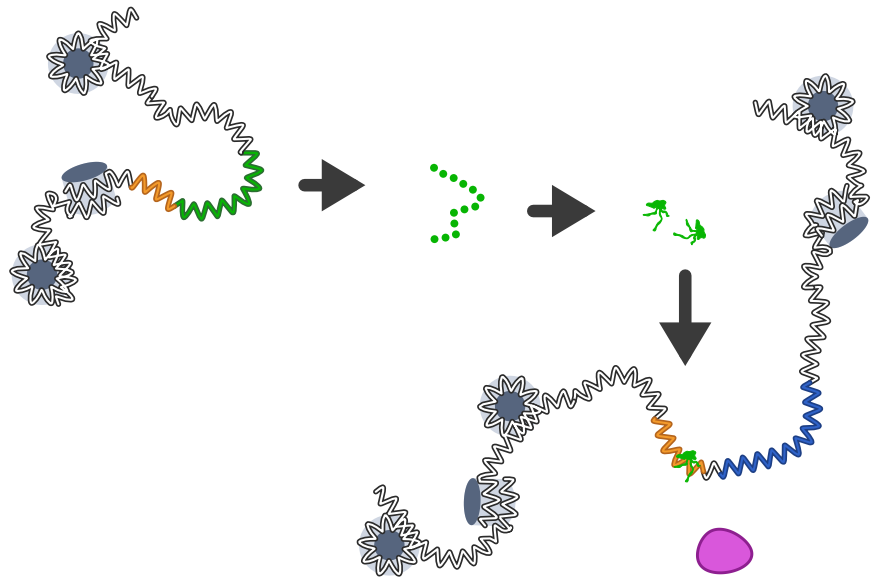


Translation

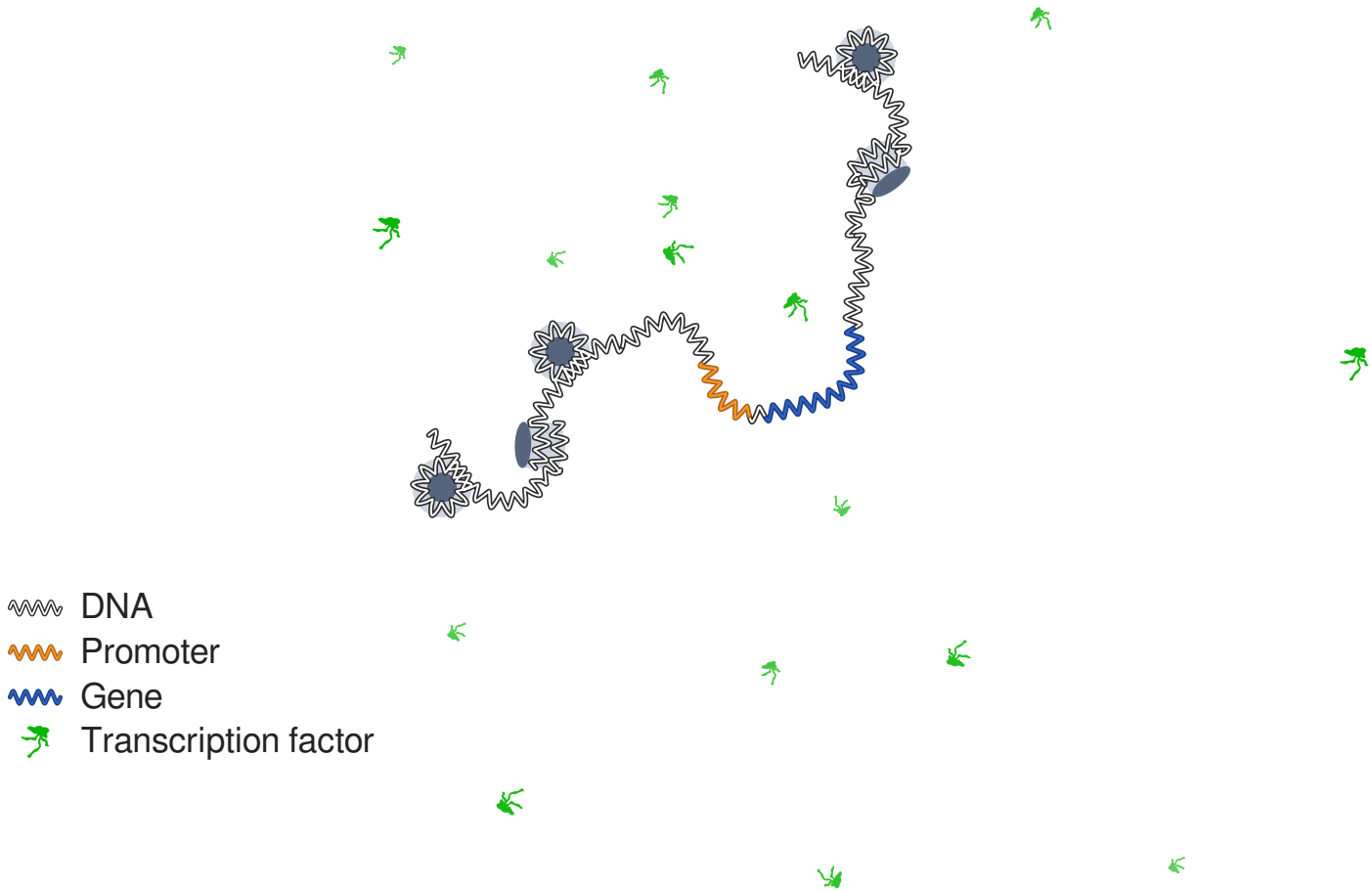
DNA
Promoter
Gene
RNA Polymerase
mRNA

Overview of gene regulation

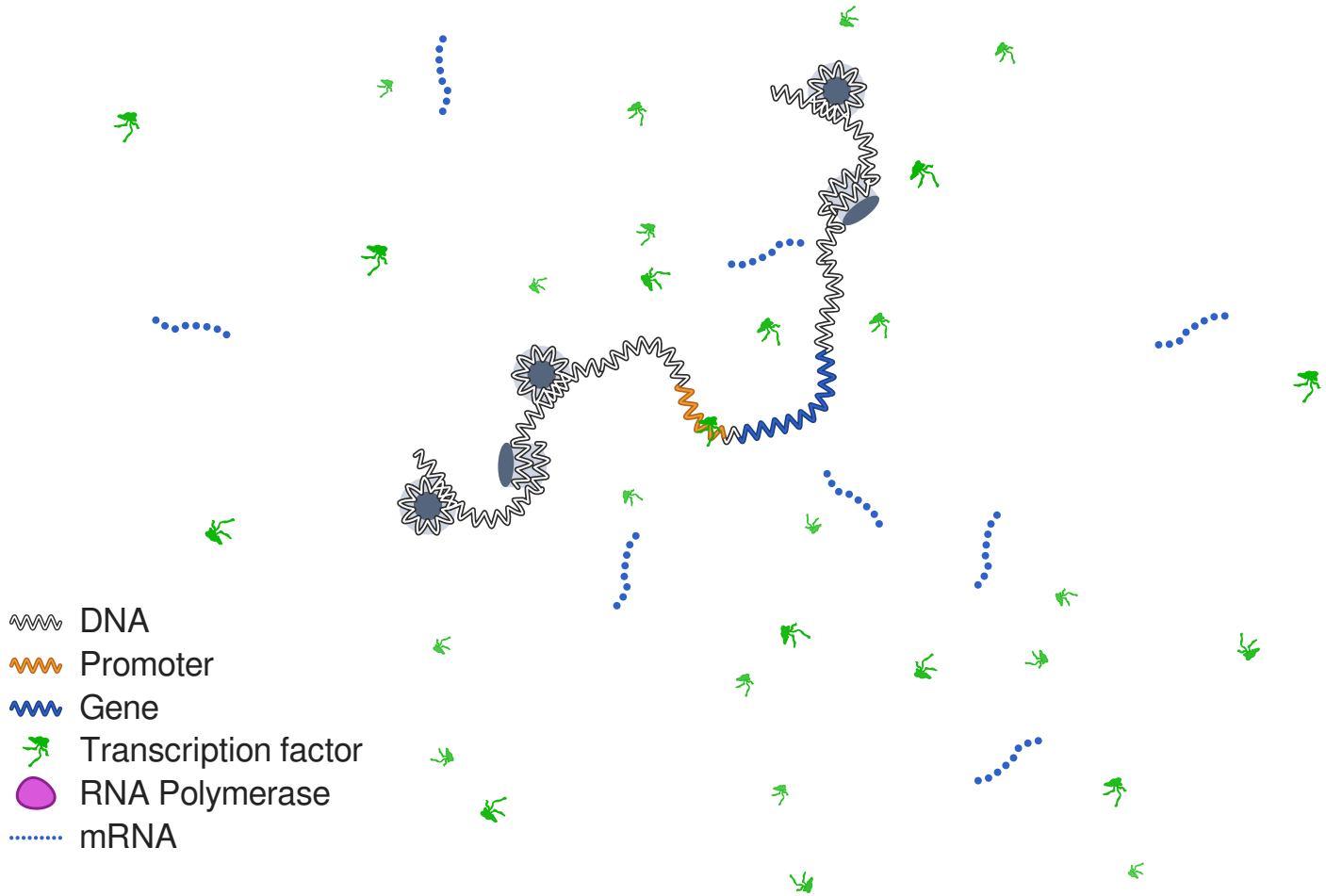


DNA
Promoter
Gene
Transcription factor
RNA Polymerase
mRNA

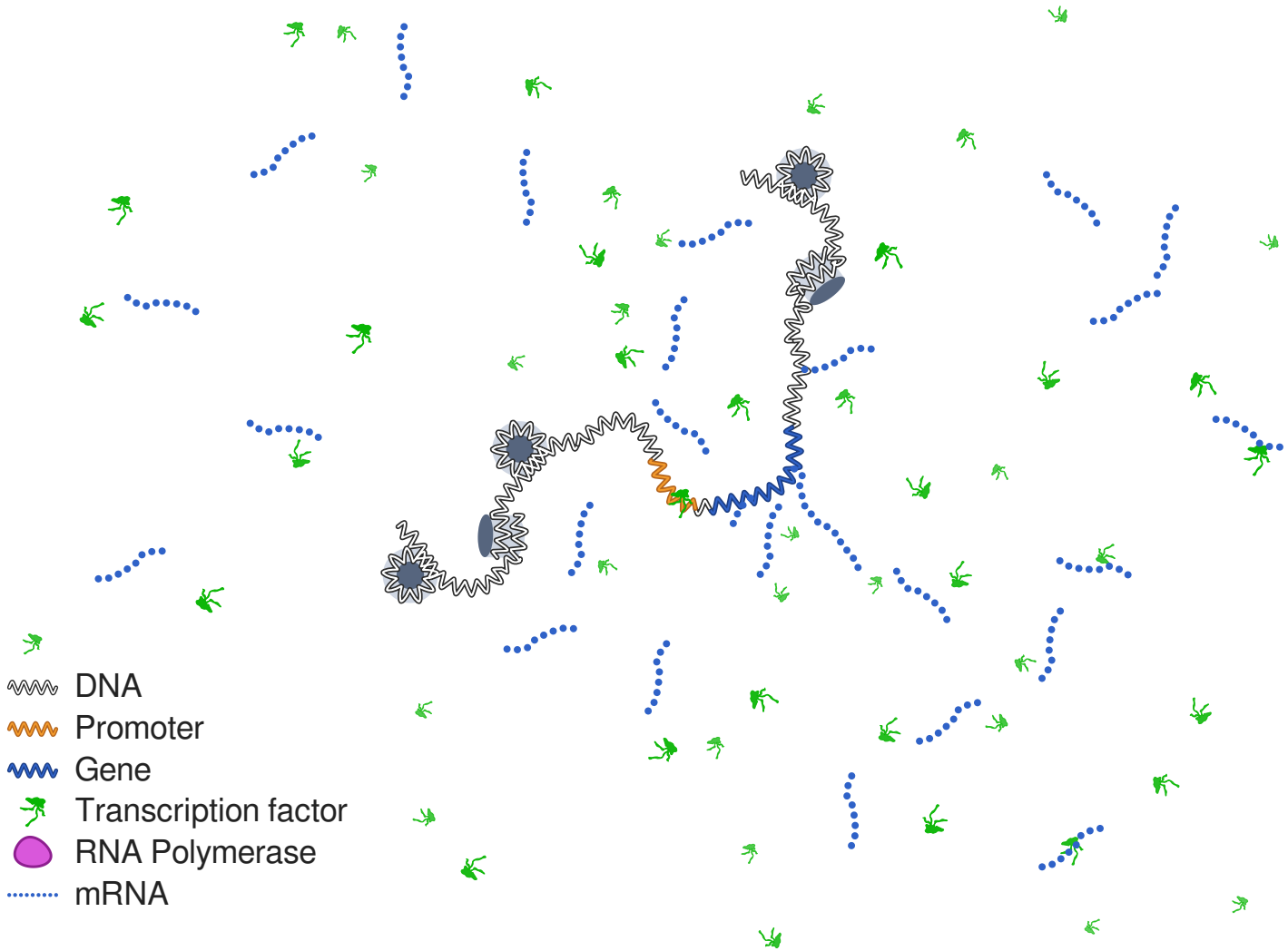
Expression levels are continuous ...



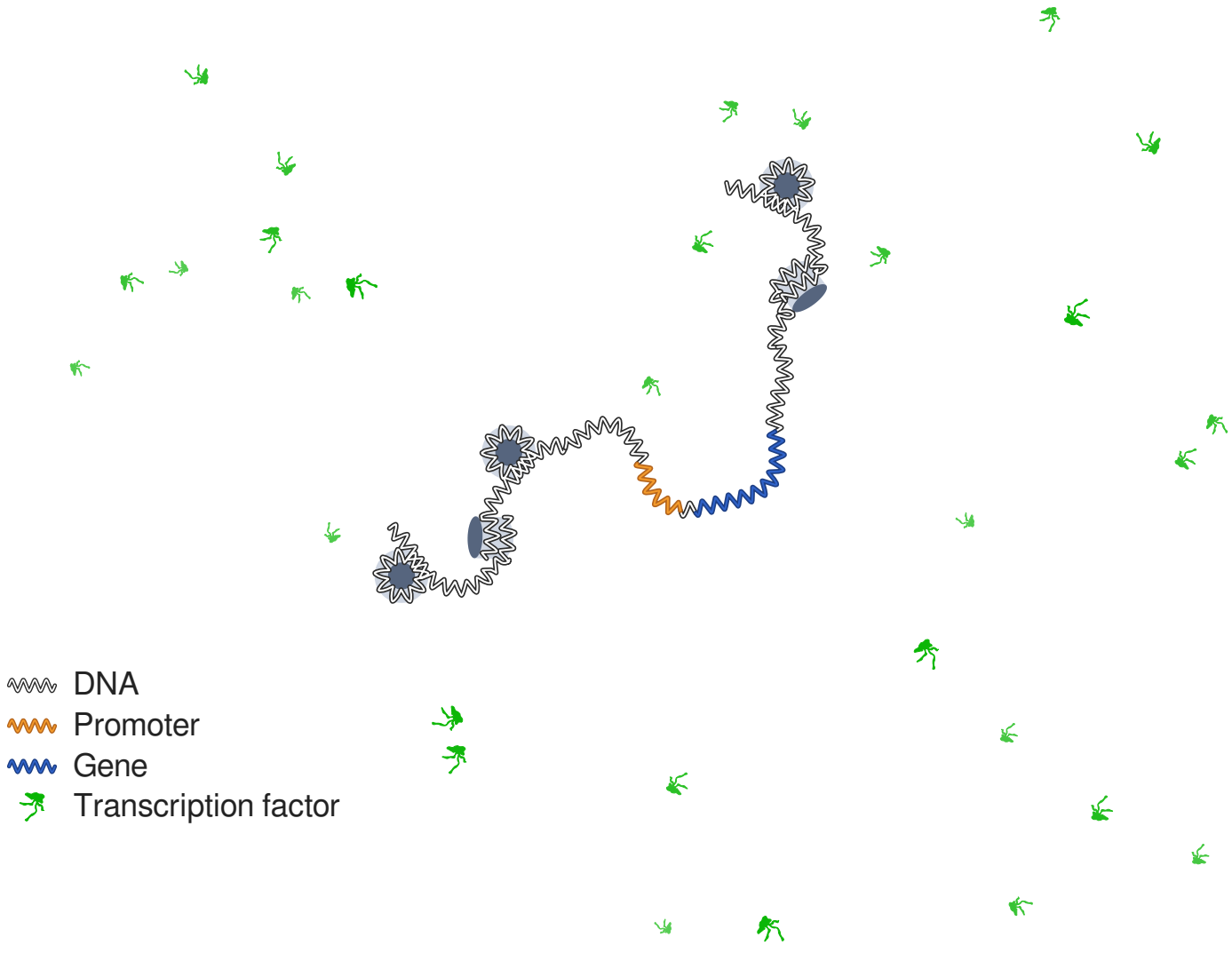
Expression levels are continuous ...



Expression levels are continuous ...



... but transcriptional control is noisy



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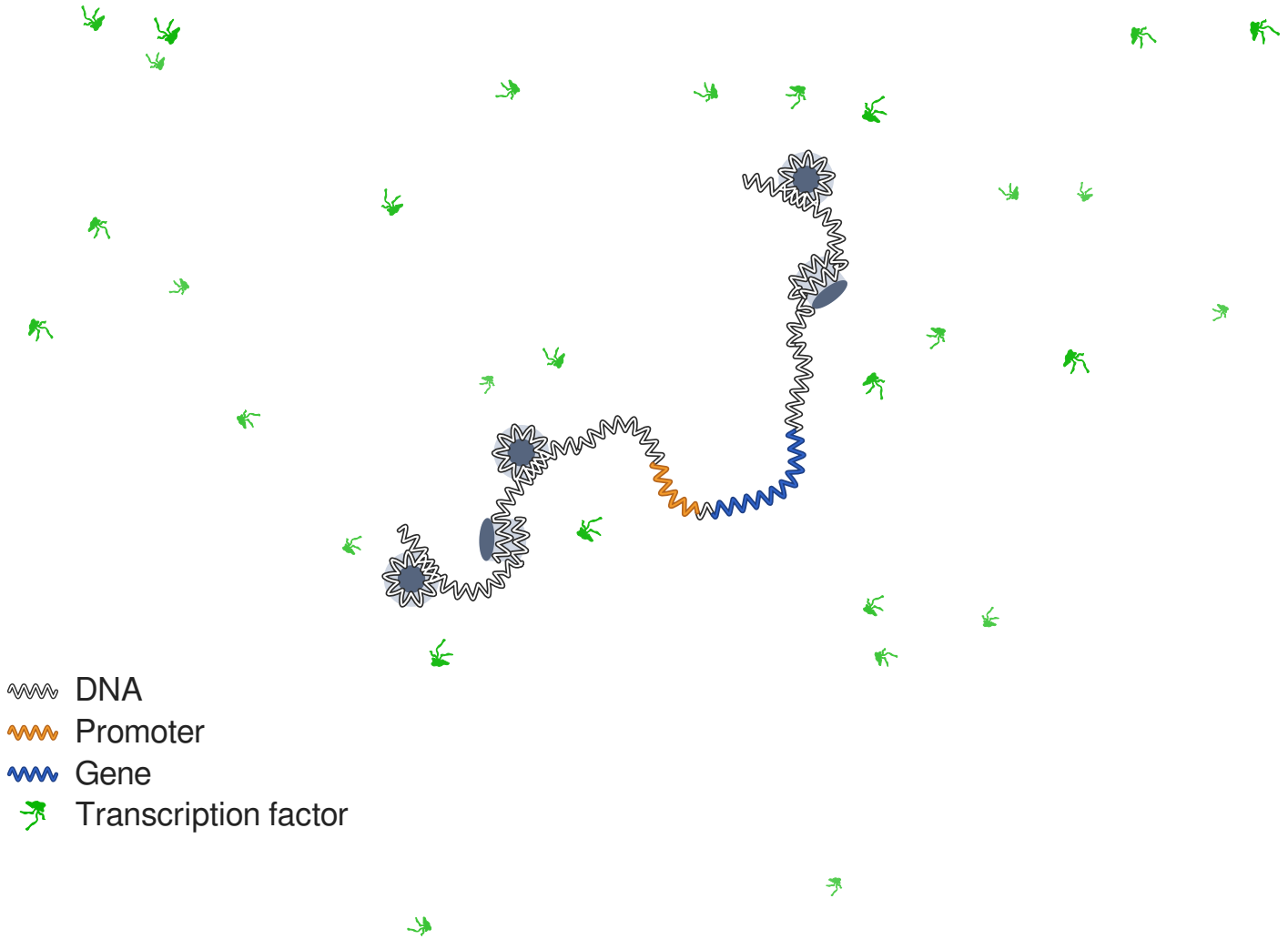

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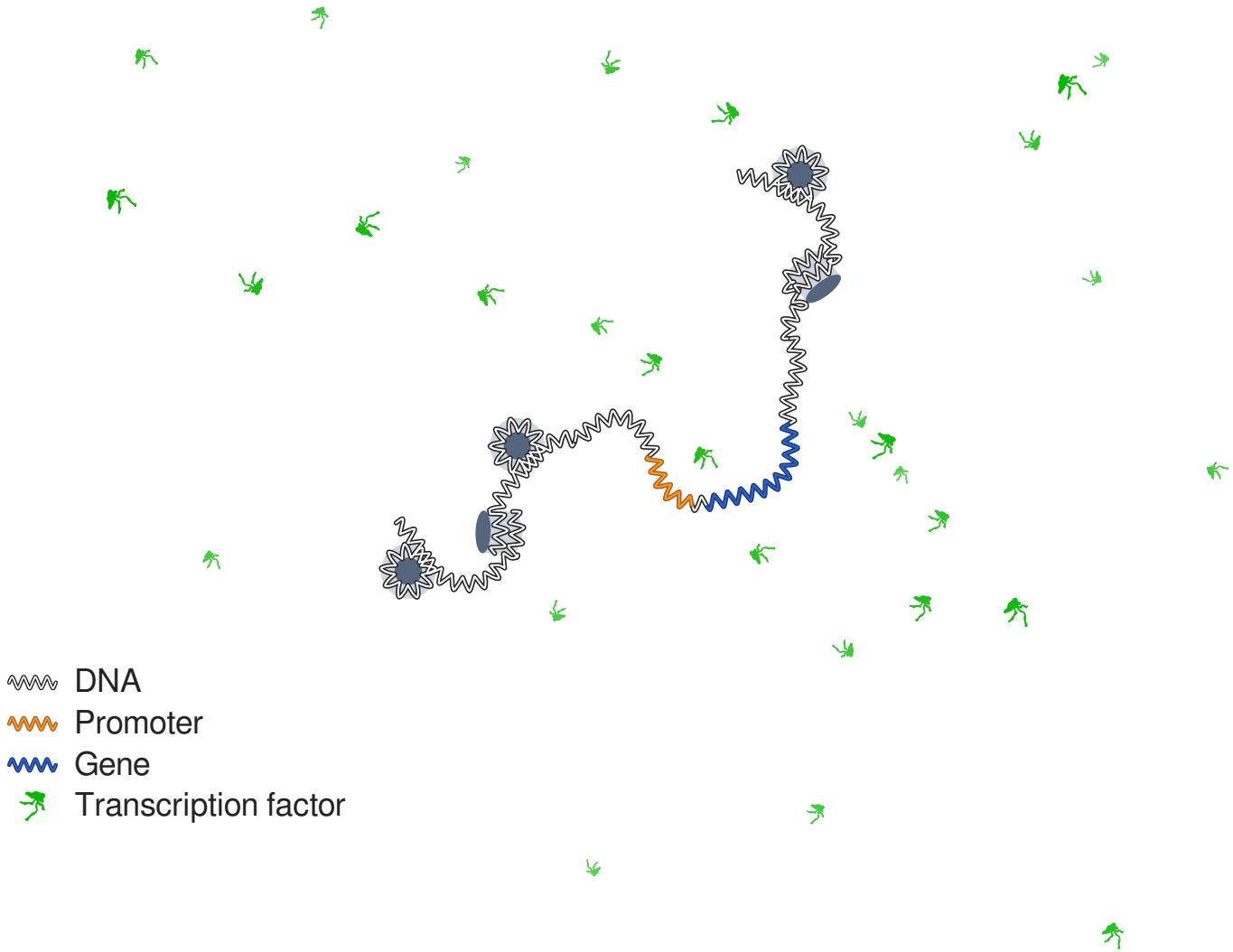

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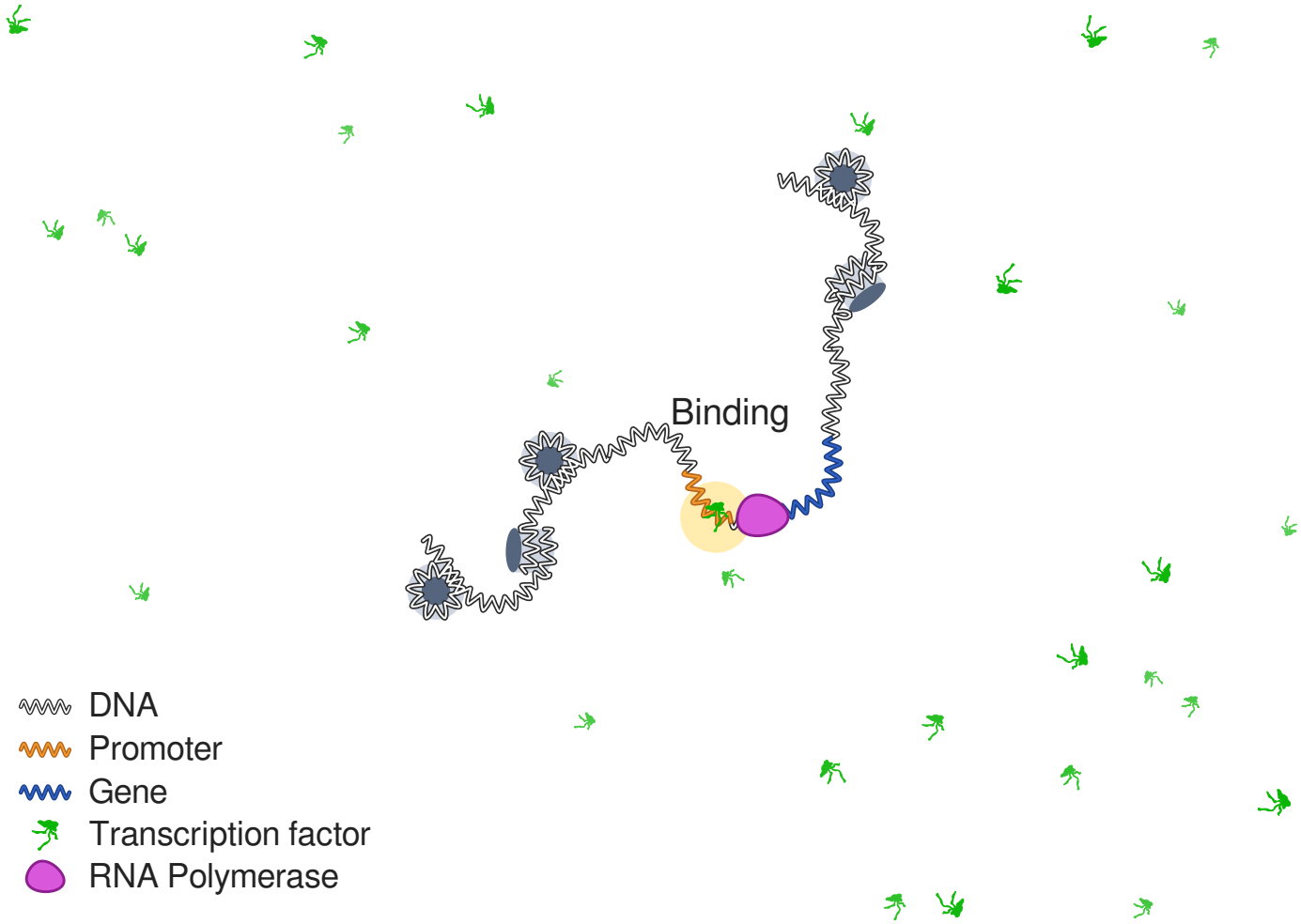

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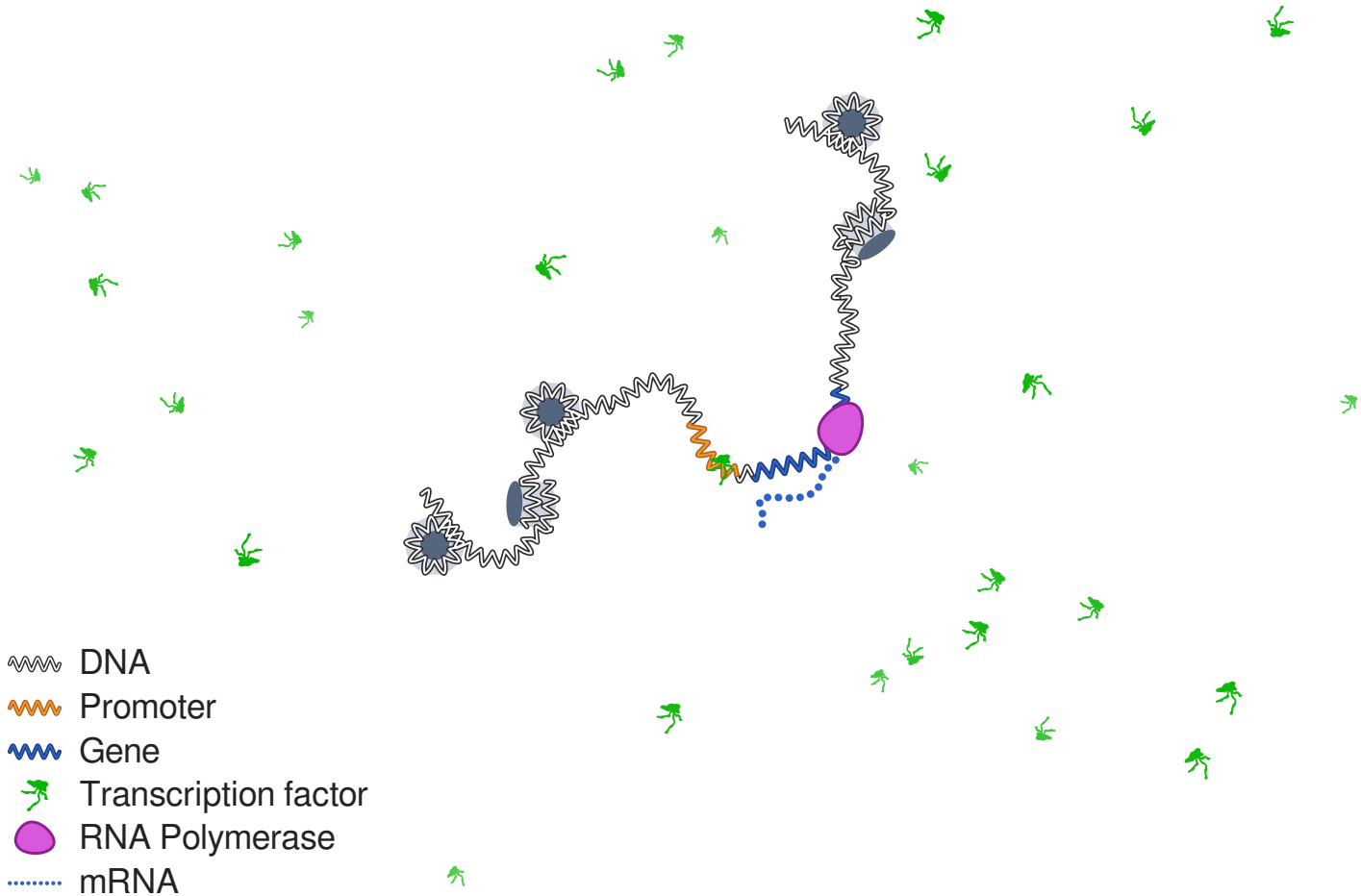
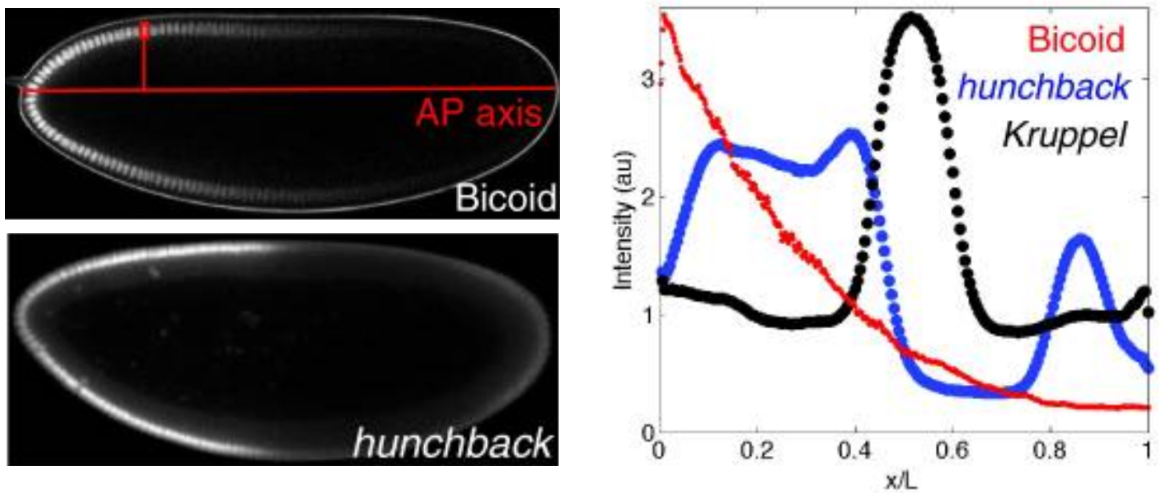
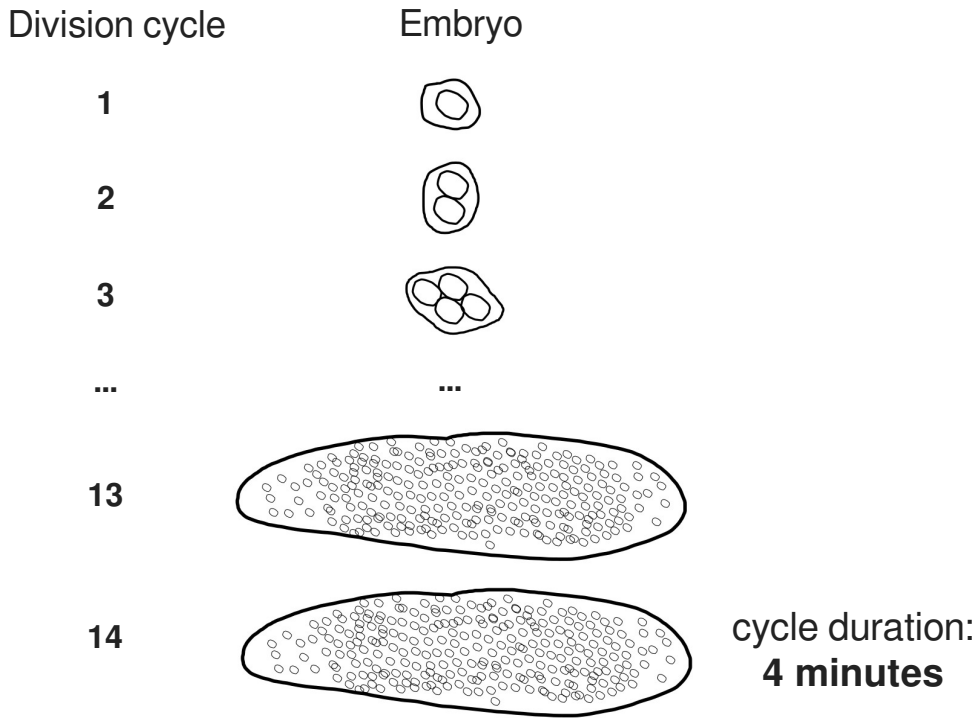


Sensitive response to shallow concentration gradients

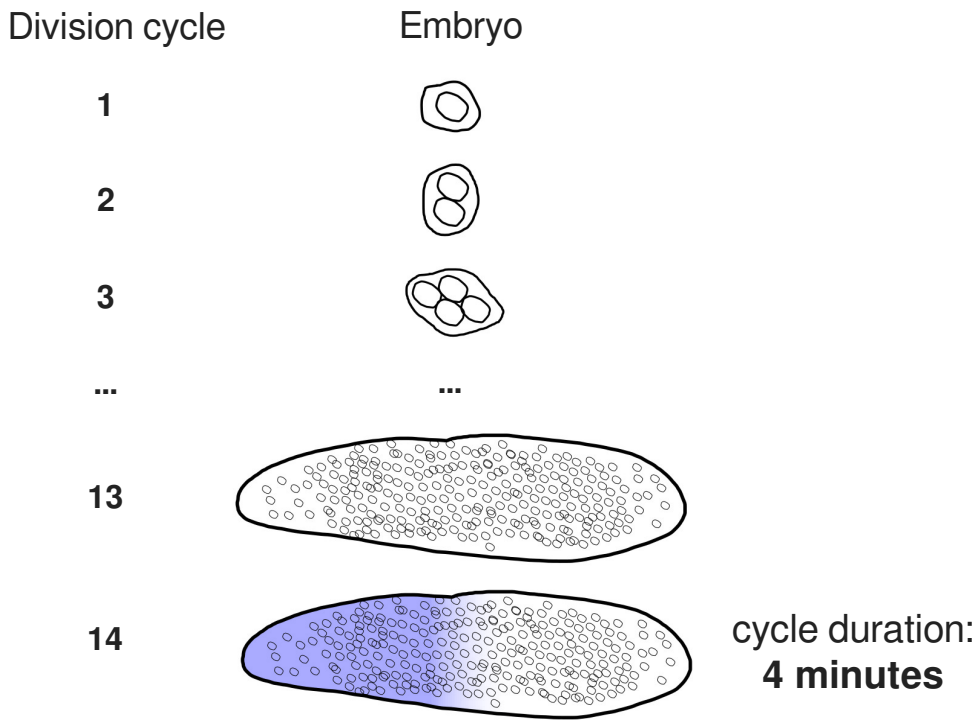


Grimm et al., *Development,* 2010

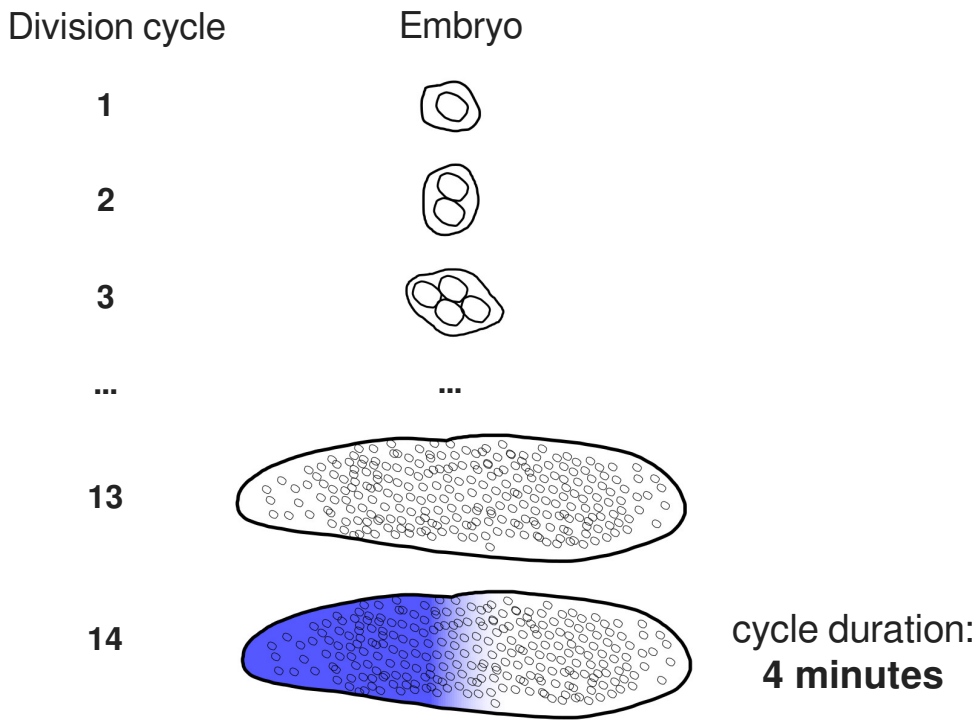
Precise sensing on a tight schedule



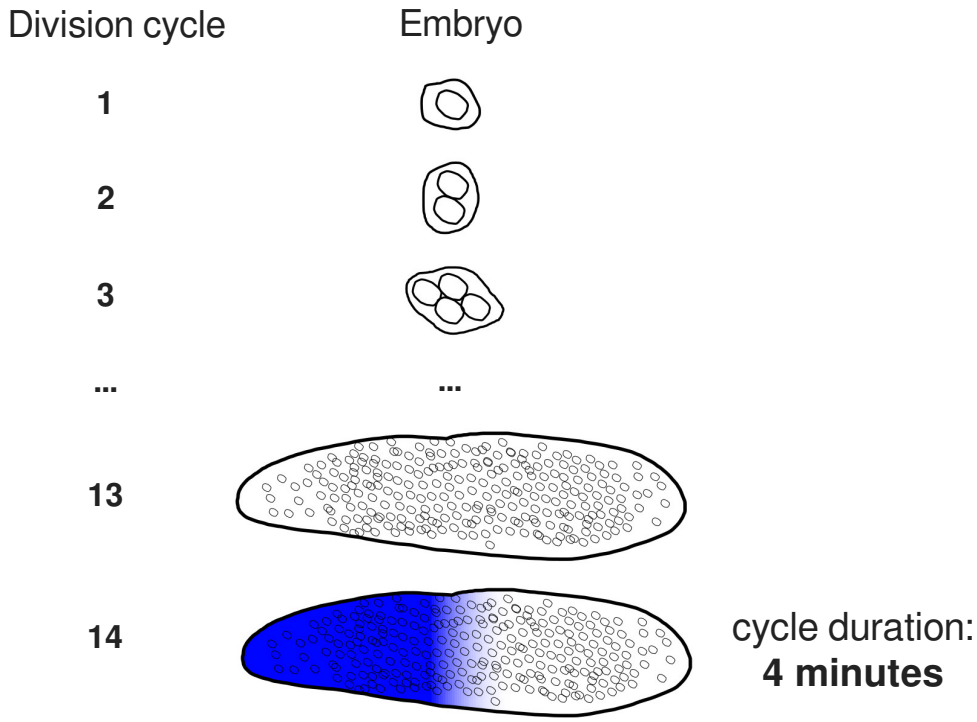
Precise sensing on a tight schedule



Precise sensing on a tight schedule



Precise sensing on a tight schedule



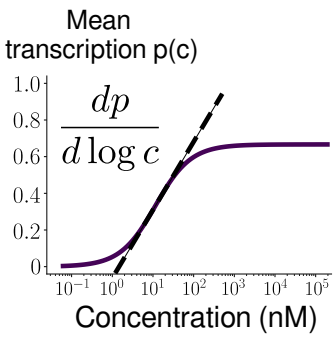
Understanding the concentration sensitivity of promoters

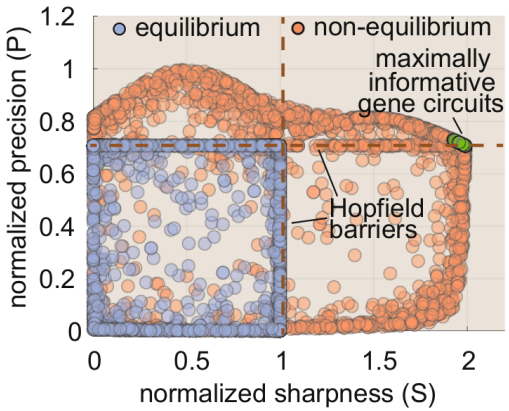
Questions:

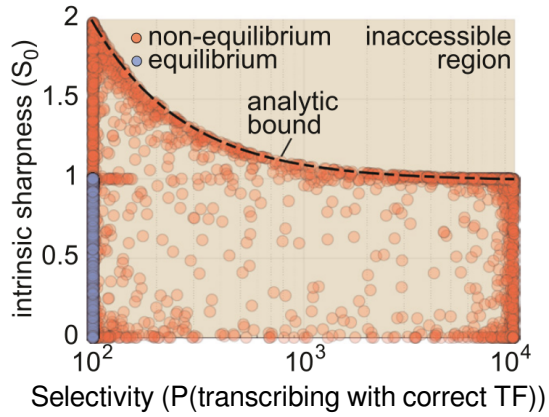
1. How to quantify sensitivity, to evaluate/compare promoter models?

2. Which promoter architectures are optimal?

3. How are these optimal architectures shaped by their constraints and by the objective function they maximize?







Lammers, Flamholz, Garcia *PNAS* 2023

Issues: 1. No accounting for physics-imposed rate constraints

2. Measure sensitivity in bits/burst

Understanding the concentration sensitivity of promoters

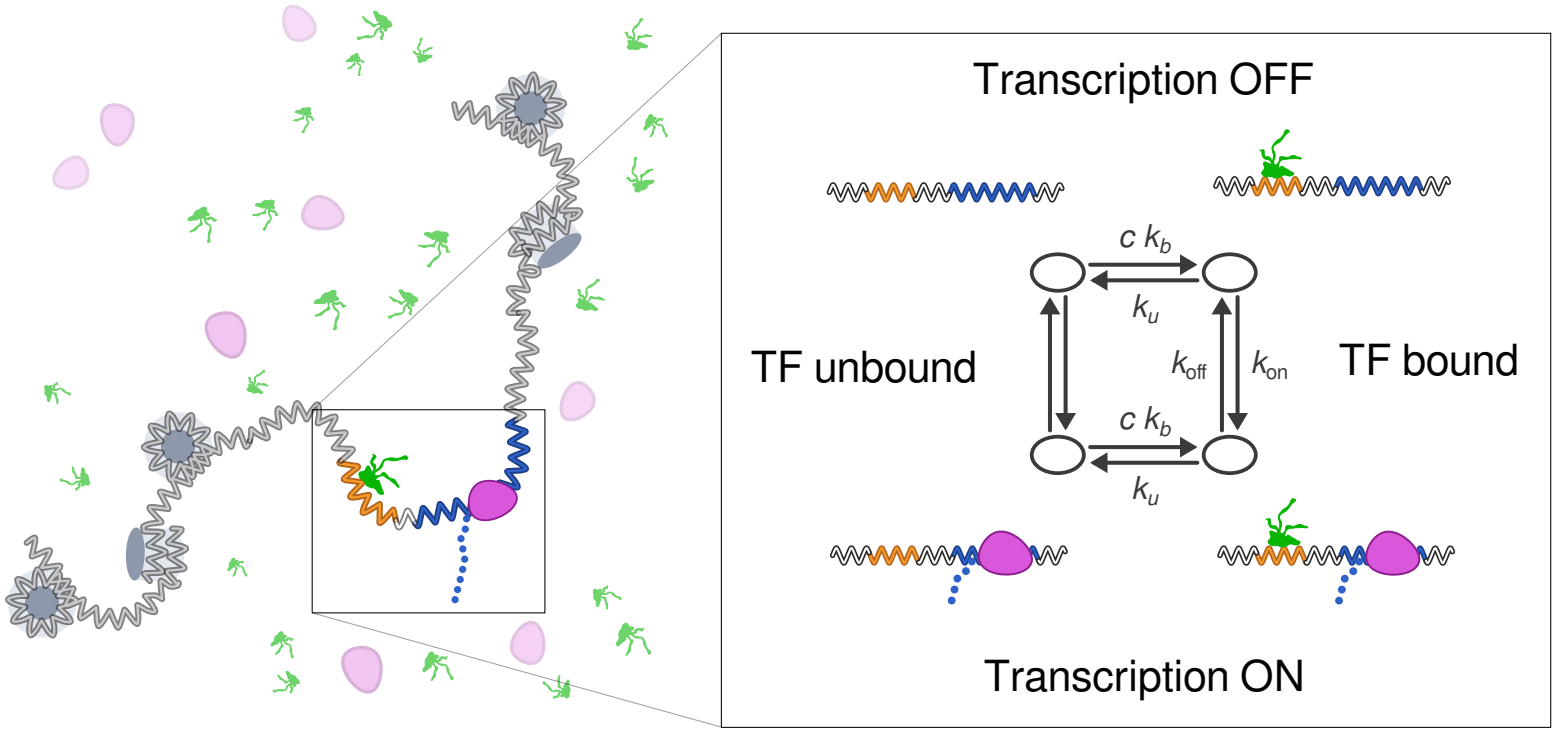
Approach:

1. Defining a measure of precision

2. Identifying physics-imposed constraints on promoter architecture

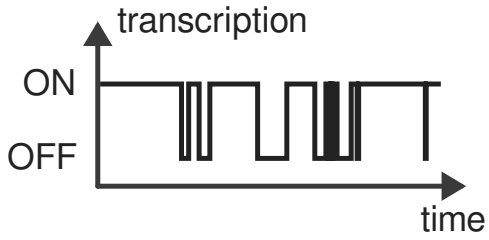
2. Constrained optimization of precision

A simple model of transcriptional control



DNA
Promoter
Gene
Transcription factor
RNA Polymerase
mRNA

Concentration: high



low

Sequential hypothesis testing

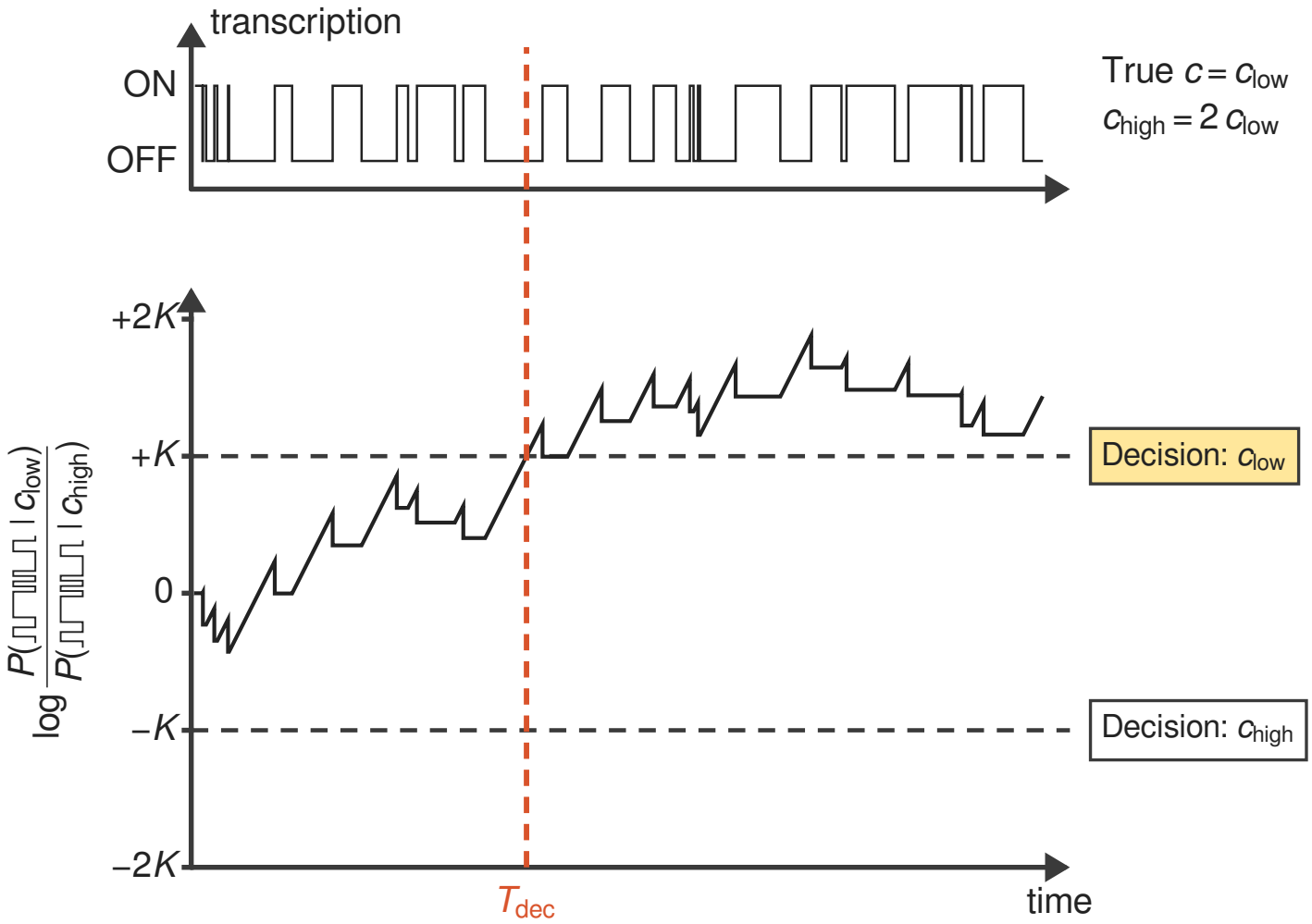
$P(\text{⎍⎍⎍⎍} \mid c_{\text{low}})$ $\qquad$ $P(\text{⎍⎍⎍⎍} \mid c_{\text{high}})$

$$R(t) = \log \frac{P(\text{⎍⎍⎍⎍} \mid c_{\text{low}})}{P(\text{⎍⎍⎍⎍} \mid c_{\text{high}})} \begin{cases} > \text{confidence value K} \rightarrow \boxed{\text{Decision: } c_{\text{low}}} \\ < -\text{K} \rightarrow \boxed{\text{Decision: } c_{\text{high}}} \end{cases}$$

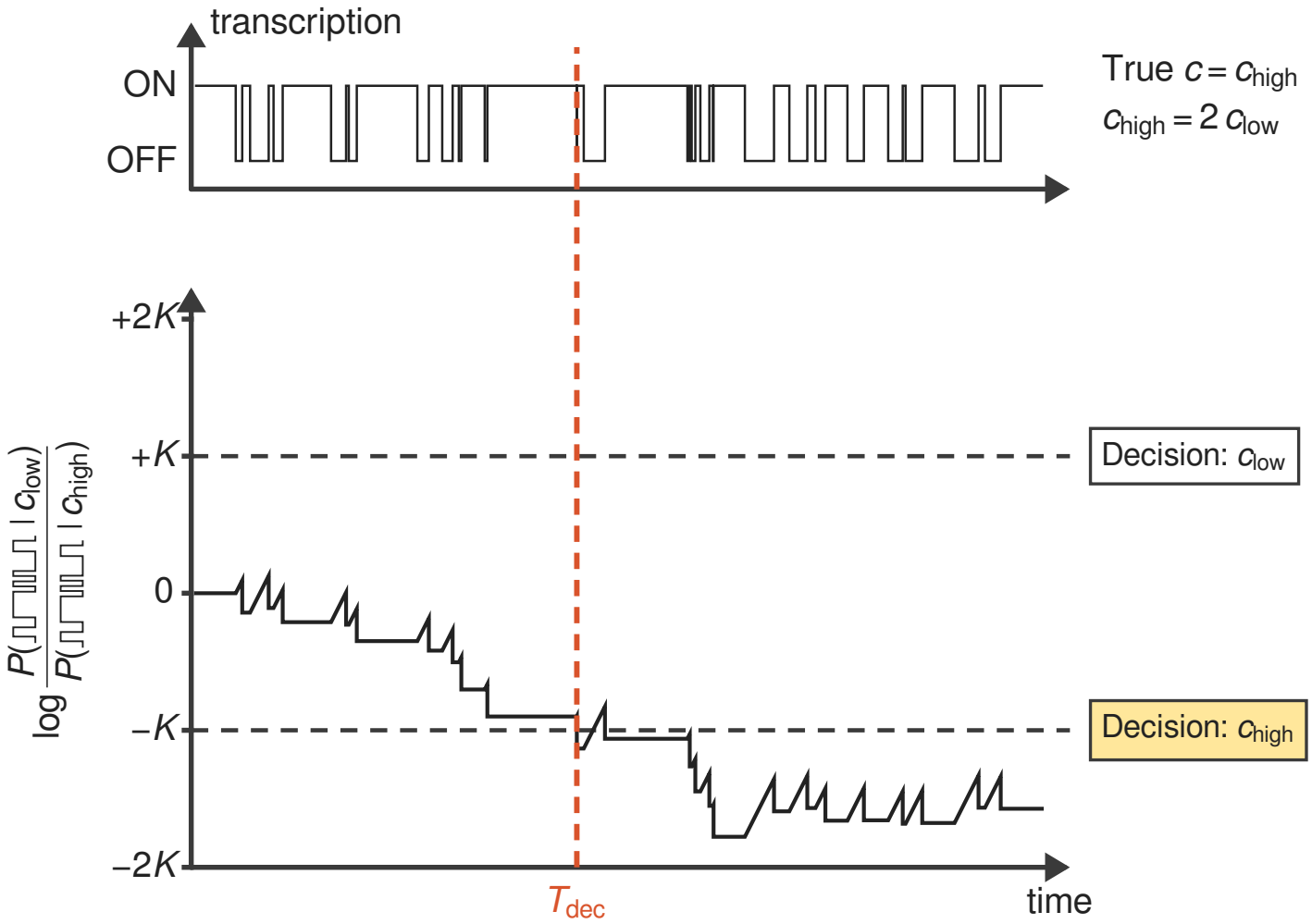
Sequential Probability Ratio Test (SPRT)

Wald 1949 (conception)
Siggia, Vergassola 2012 (application to biological receptors)
Desponds, Vergassola, Walczak 2022 (optimization)
Lammers, Flamholz, Garcia, 2024 (nonequilibrium receptors)

Sequential hypothesis testing



Sequential hypothesis testing



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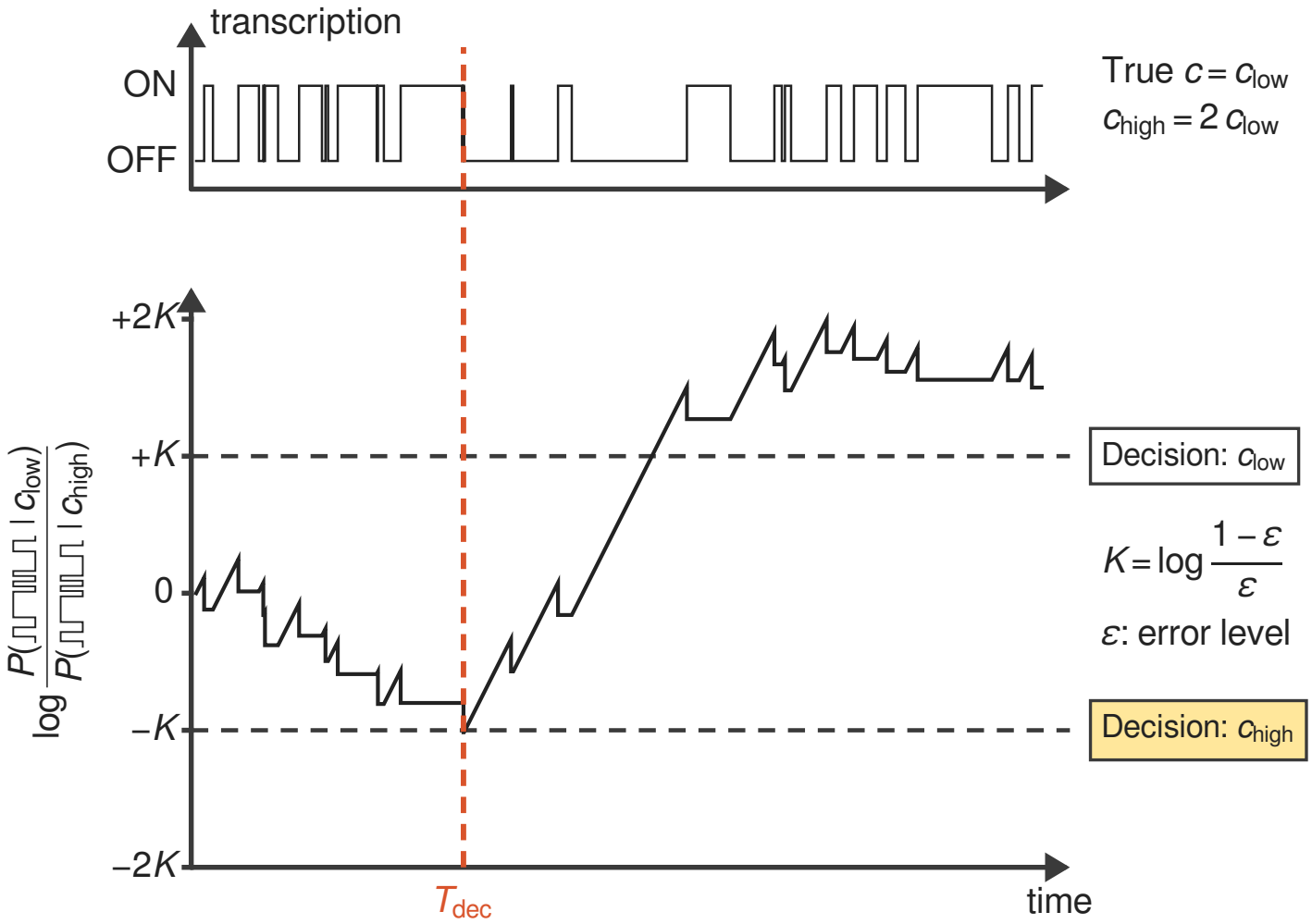
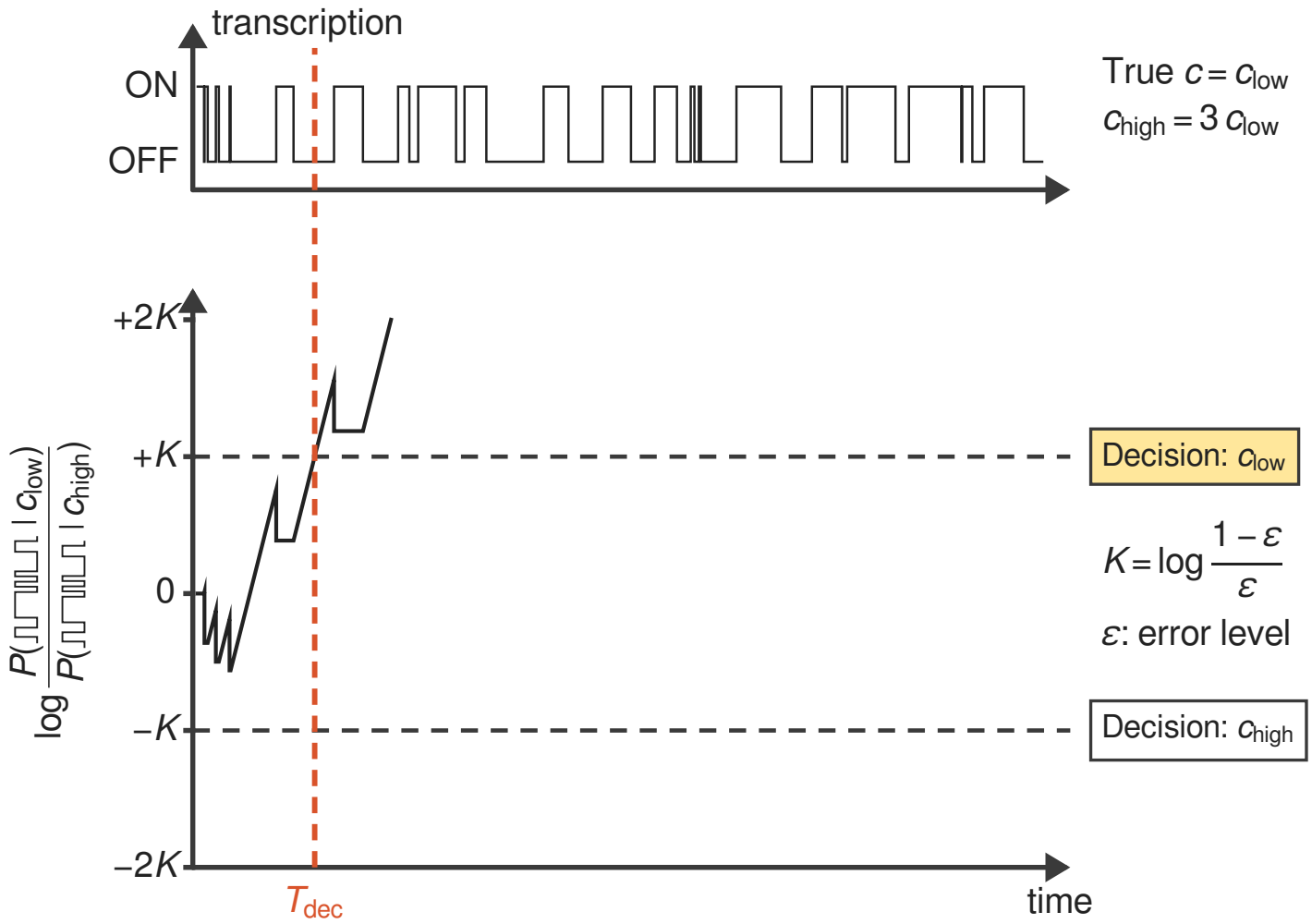


Sequential hypothesis testing



Fisher information rate and decision time

Drift-diffusion approximation when $\delta c/c \ll 1$

$$\frac{d\mathcal{R}}{dt} = V + \sqrt{2V}\eta \quad \leftarrow \quad \text{Gaussian noise } \langle \eta(t)\eta(t') \rangle \sim \delta(t-t')$$

V = Drift, $\sqrt{2V}$ = Diffusion

SPRT mean decision time (Siggia & Vergassola 2012)

$$\langle T_{\text{dec}} \rangle = \frac{K}{V} \tanh\left(\frac{K}{2}\right)$$

$$V \approx \frac{1}{2}\left(\frac{\delta c}{c}\right)^2 \dot{\mathcal{F}}(c)$$

$$\dot{\mathcal{F}}(c) = -\lim_{t\to\infty} \frac{1}{t} \left\langle \frac{\partial^2 \mathcal{L}(c)}{(\partial \ln c)^2} \right\rangle_c$$

$$\mathcal{L}(c;t) = \ln P(\text{ } \,|c)$$

The optimization problem

$$\text{Maximize } \dot{\mathcal{F}}(c)$$

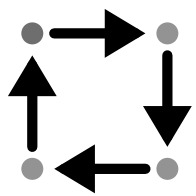
subject to the constraints

$$k_b \leq 4aD \text{ and } k_{\text{u,on,off}} \leq k_{\max}$$

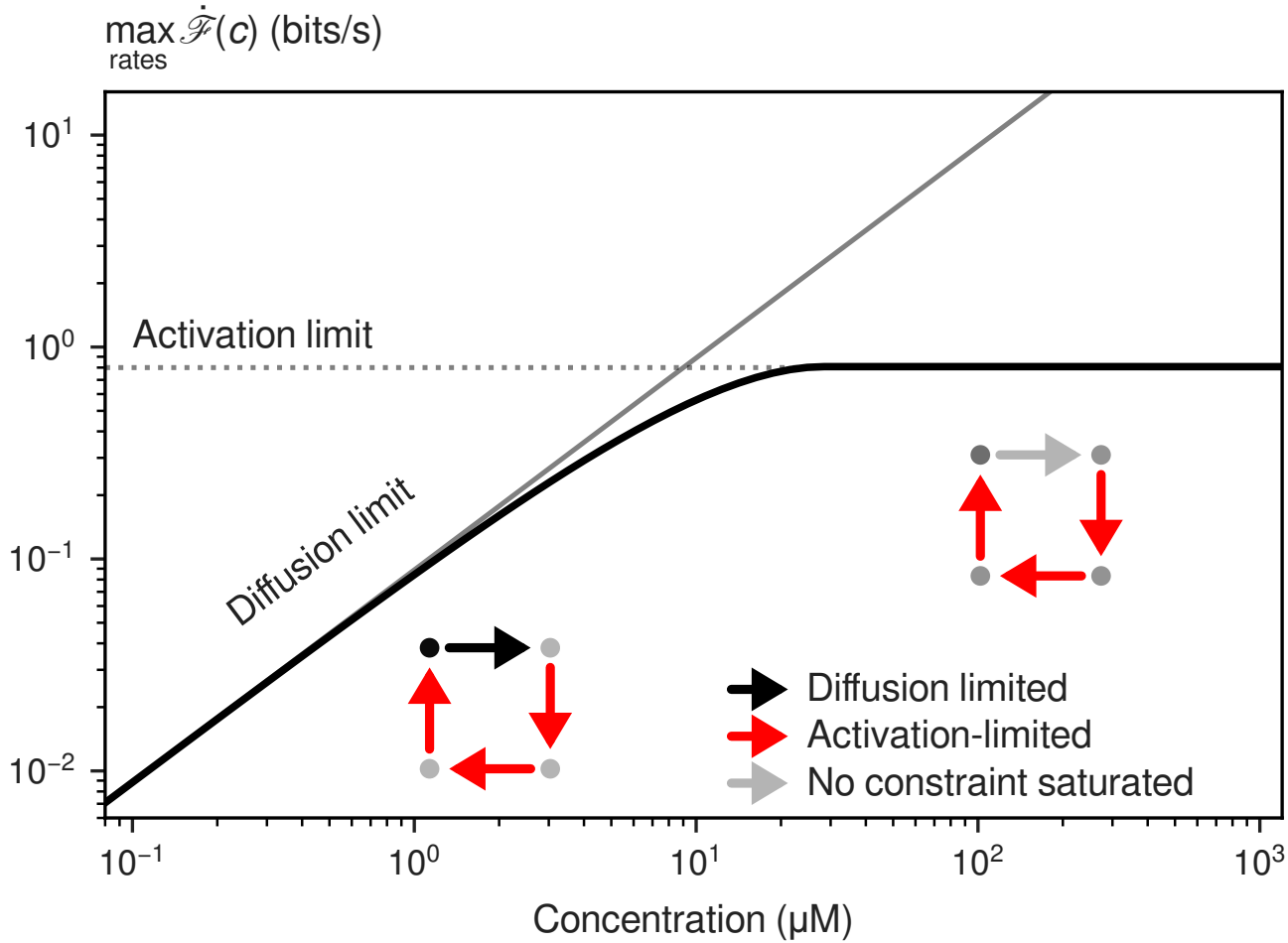
Diffusion Activation

In practice:

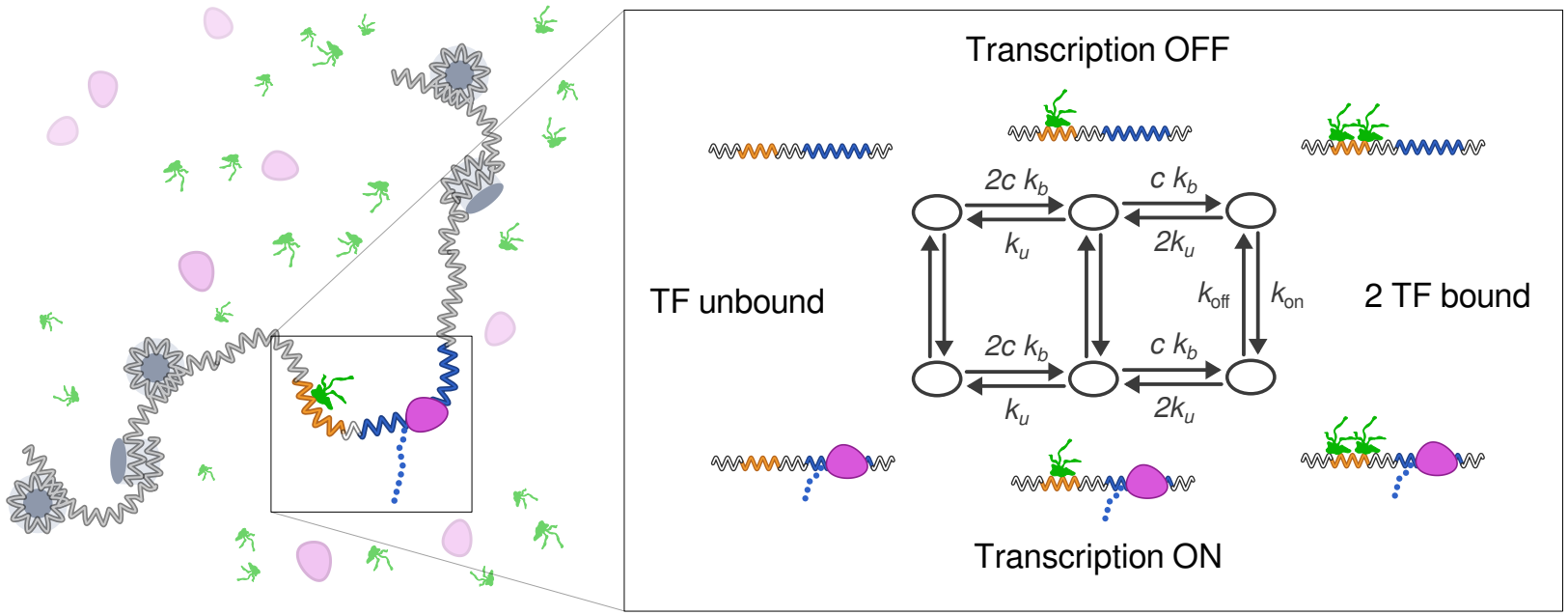
1. numerical first pass (dynamic programming)
2. ansatz obtained from solutions (analytical)



Optimal promoters: two distinct regimes



Cooperative binding



DNA
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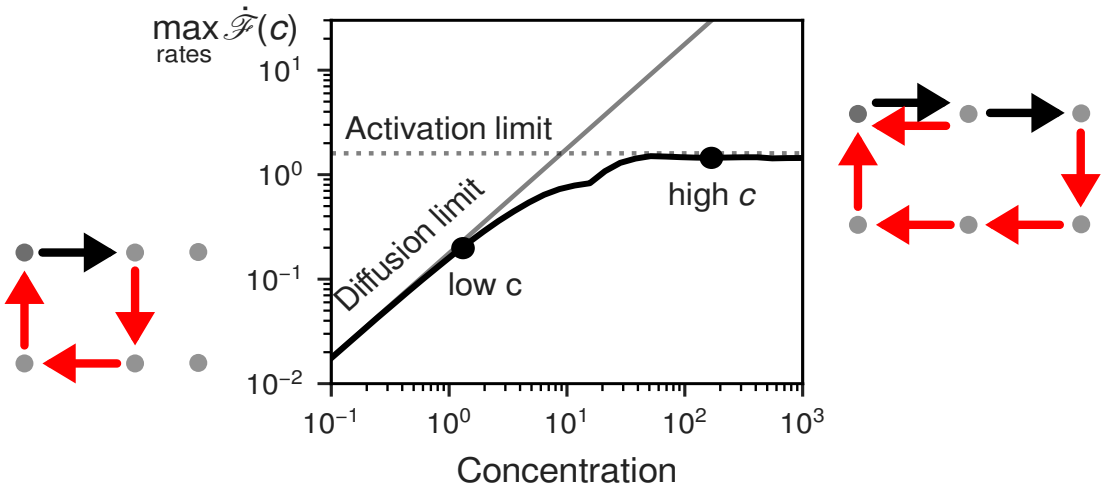
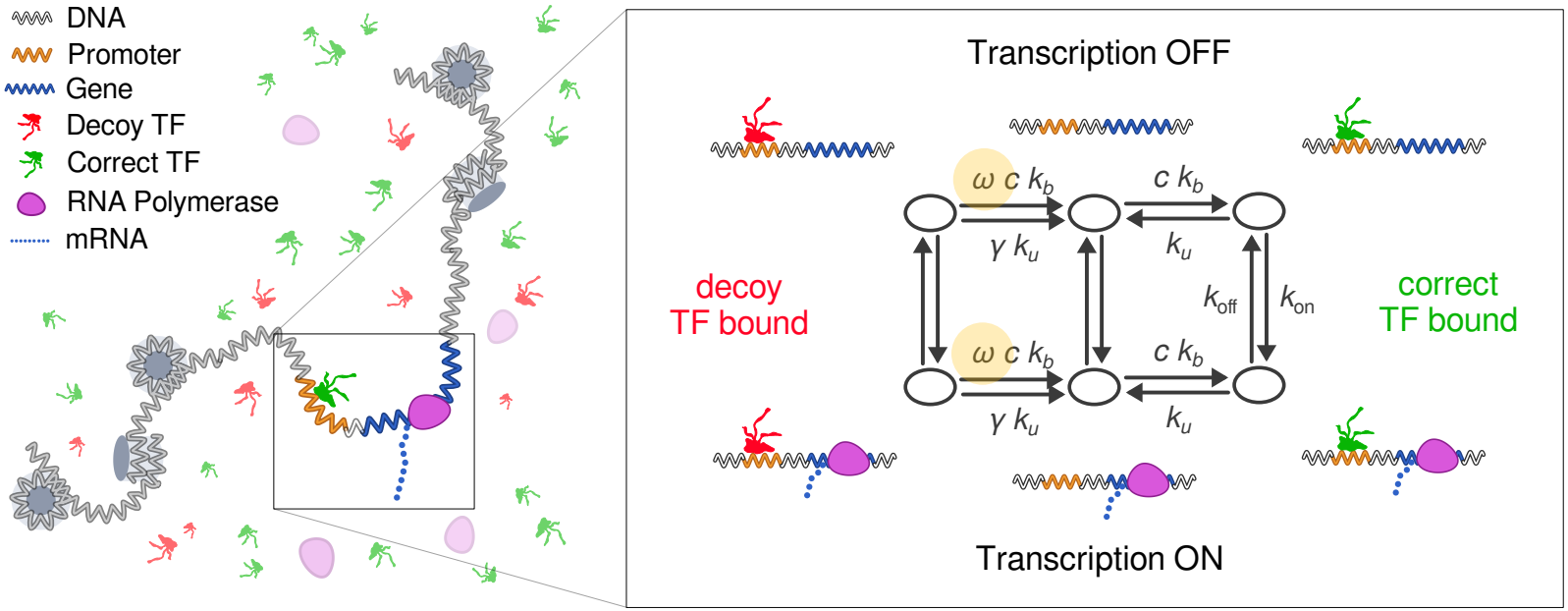
Transcription factor
RNA Polymerase
mRNA

Diffusion limited
Activation-limited
No constraint saturated

↓ Maximize $\dot{\mathcal{F}}(c)$ subject to the constraints ↓



Spurious binding



$\downarrow$ Maximize $\dot{\mathcal{F}}(c)$ subject to the constraints $\downarrow$

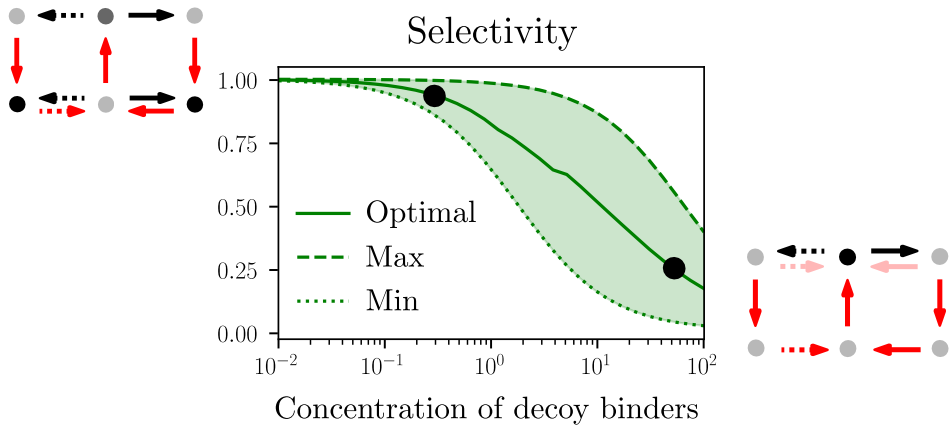
ω : [decoy TF] / [correct TF]

$$\gamma : \frac{\text{decoy unbinding rate}}{\text{correct unbinding rate}}$$

Selectivity f = P(transcribing | correct TF)

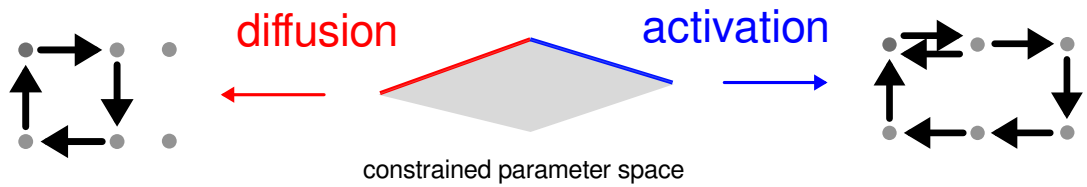
$$\max f = \gamma^2 / (\omega + \gamma^2)$$

$$\min f = \gamma / (\omega(\gamma+1)/2 + \gamma)$$

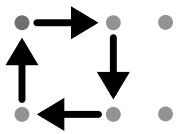


Takeaways

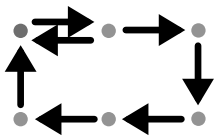
The limiting constraint shapes the optimal architecture



Solution set is disconnected: each cluster points to a different sensing strategy

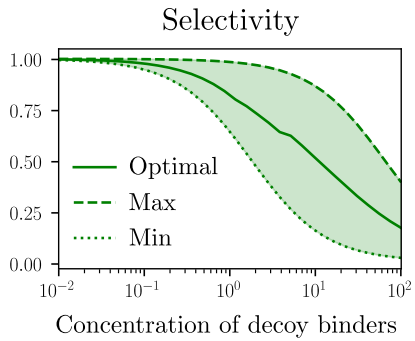


count rare
binding events

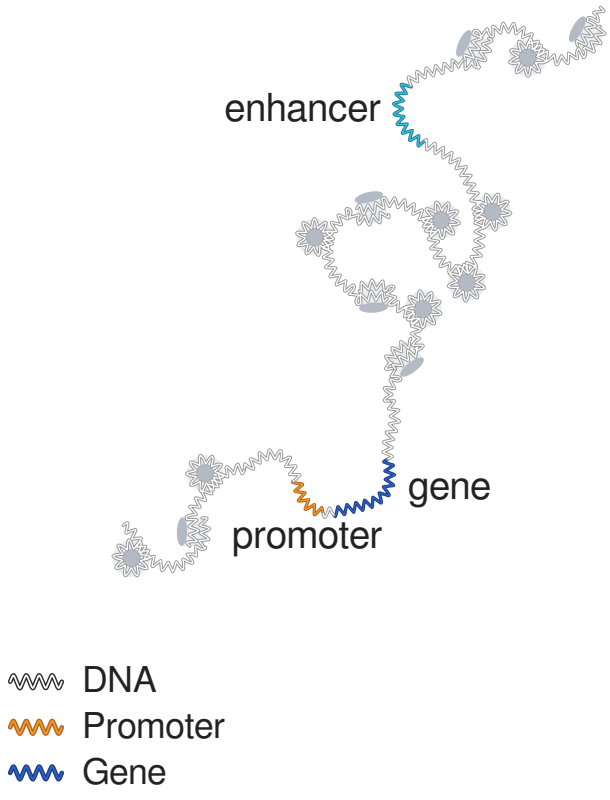


leverage
cooperative binding

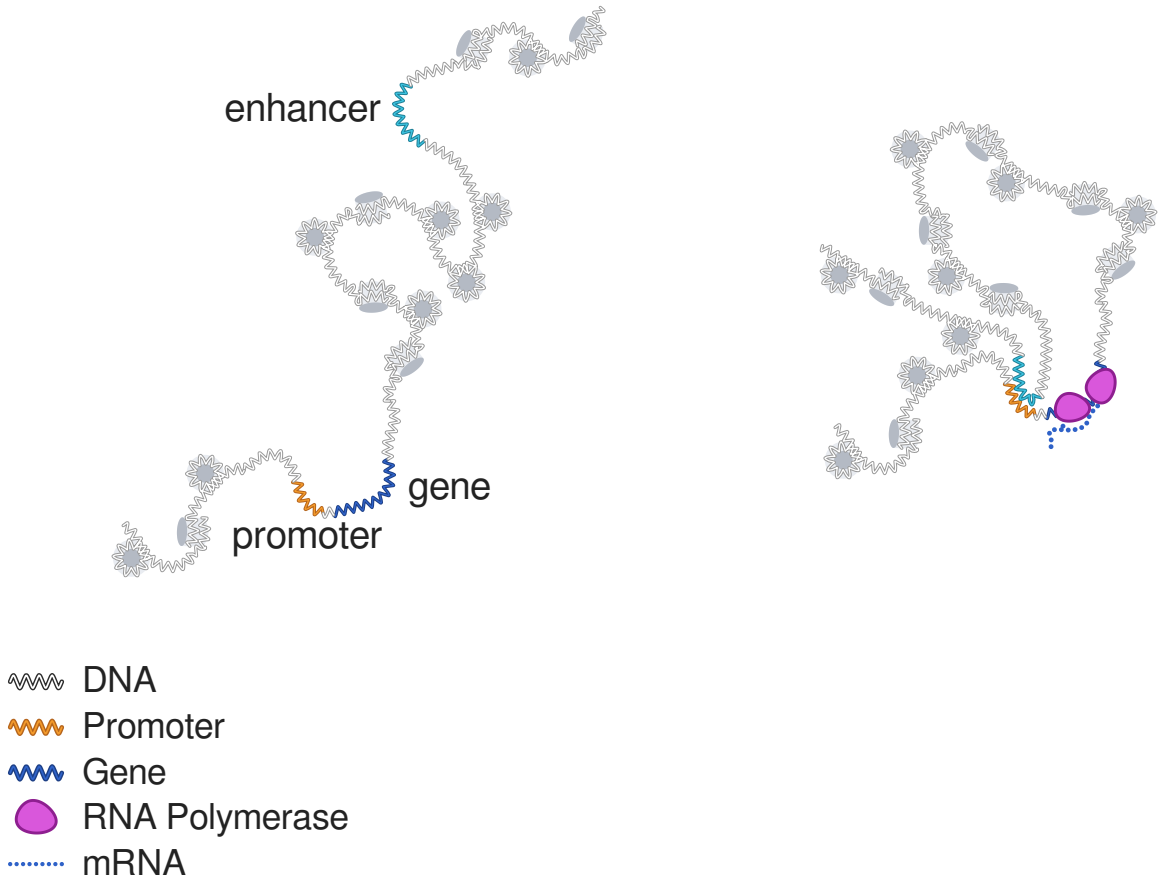
Sharpness, selectivity do not lie at extrema



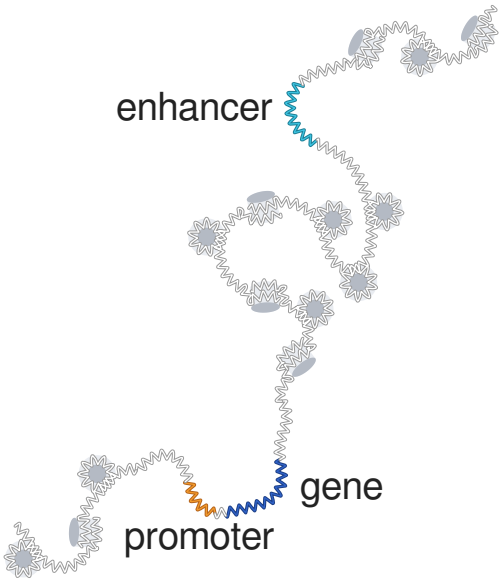
Transcriptional control by distal elements

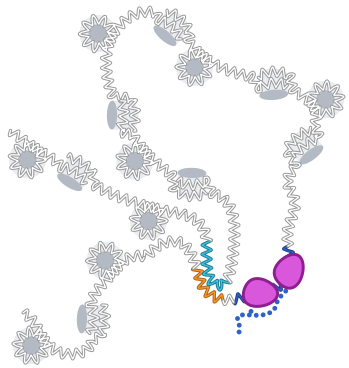


Transcriptional control by distal elements



Transcriptional control by distal elements





DNA
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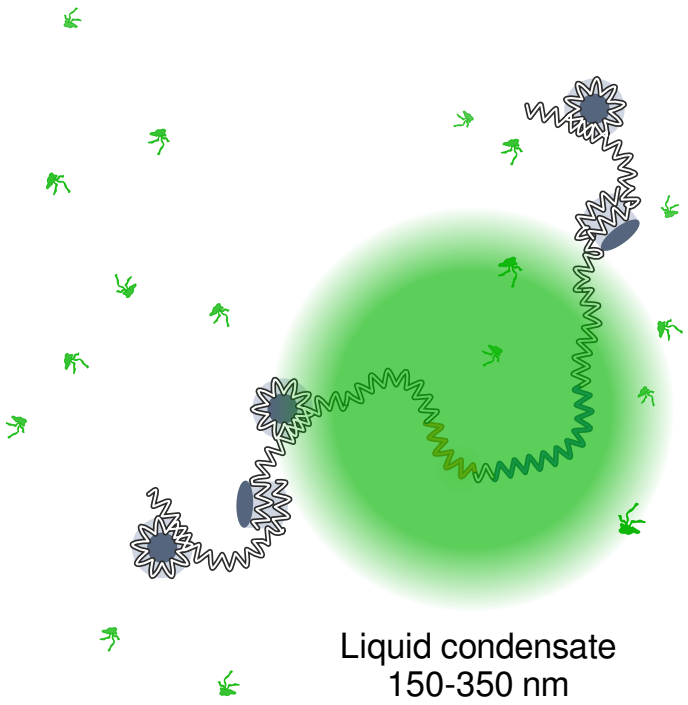
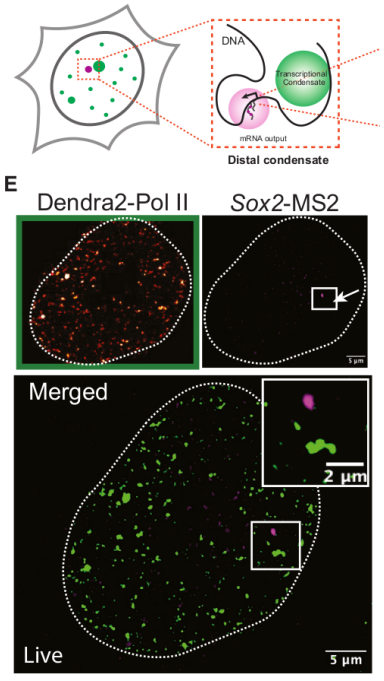
Gene
RNA Polymerase
mRNA

Fukaya et al., *Cell,* 2016
Chen et al., *Nature Genetics,* 2018
Brückner et al., *Science,* 2023

Transcriptional control through liquid phase separation?



Liquid condensate
150-350 nm



Du et al.,*Cell* 2024

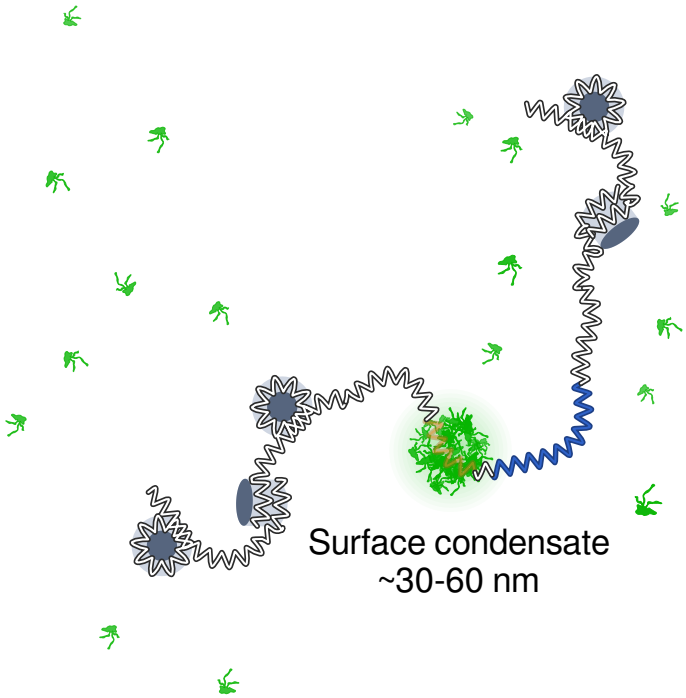
- DNA
- Promoter
- Gene
- Transcription factor

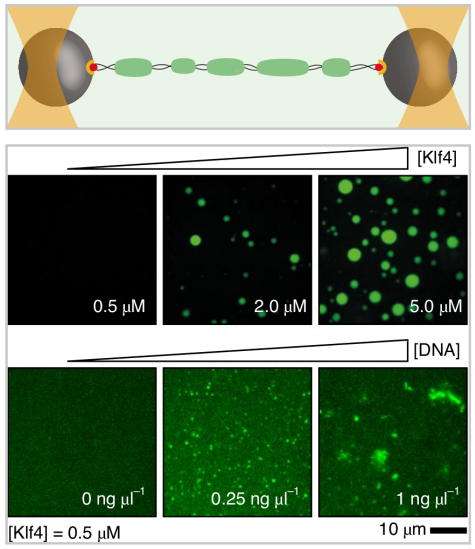
Hnisz et al., *Cell* 2017

Sabari et al., *Science* 2018

Munshi et al., *Science Advances* 2025

Proteins condense on the surface of DNA





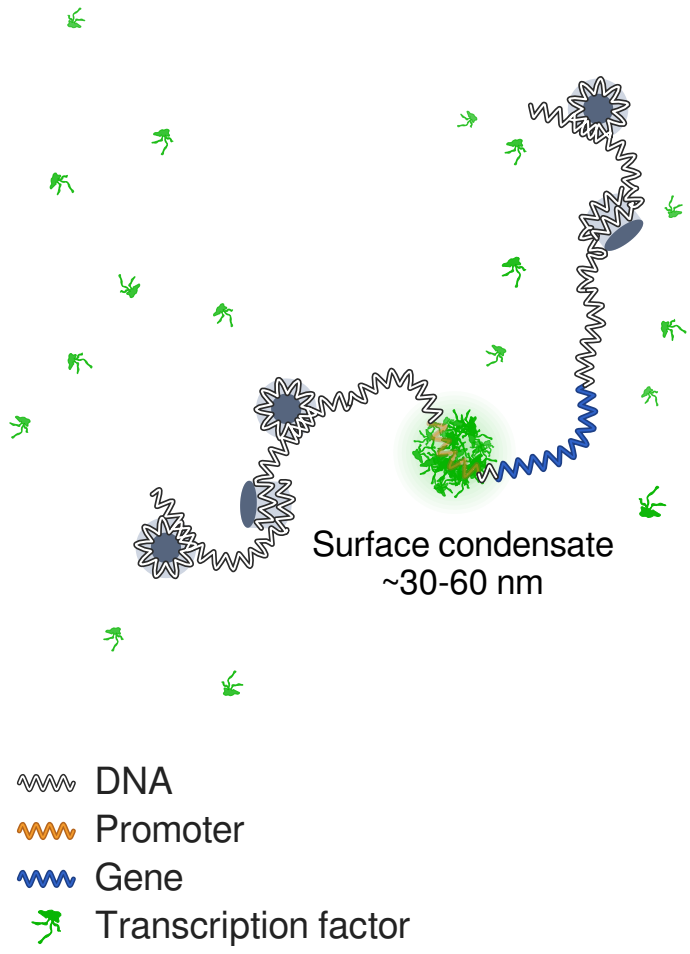
Morin et al., *Nature Physics* 2022

DNA
Promoter
Gene
Transcription factor

Pancholi et al., *MolecularSystems Biology* 2021

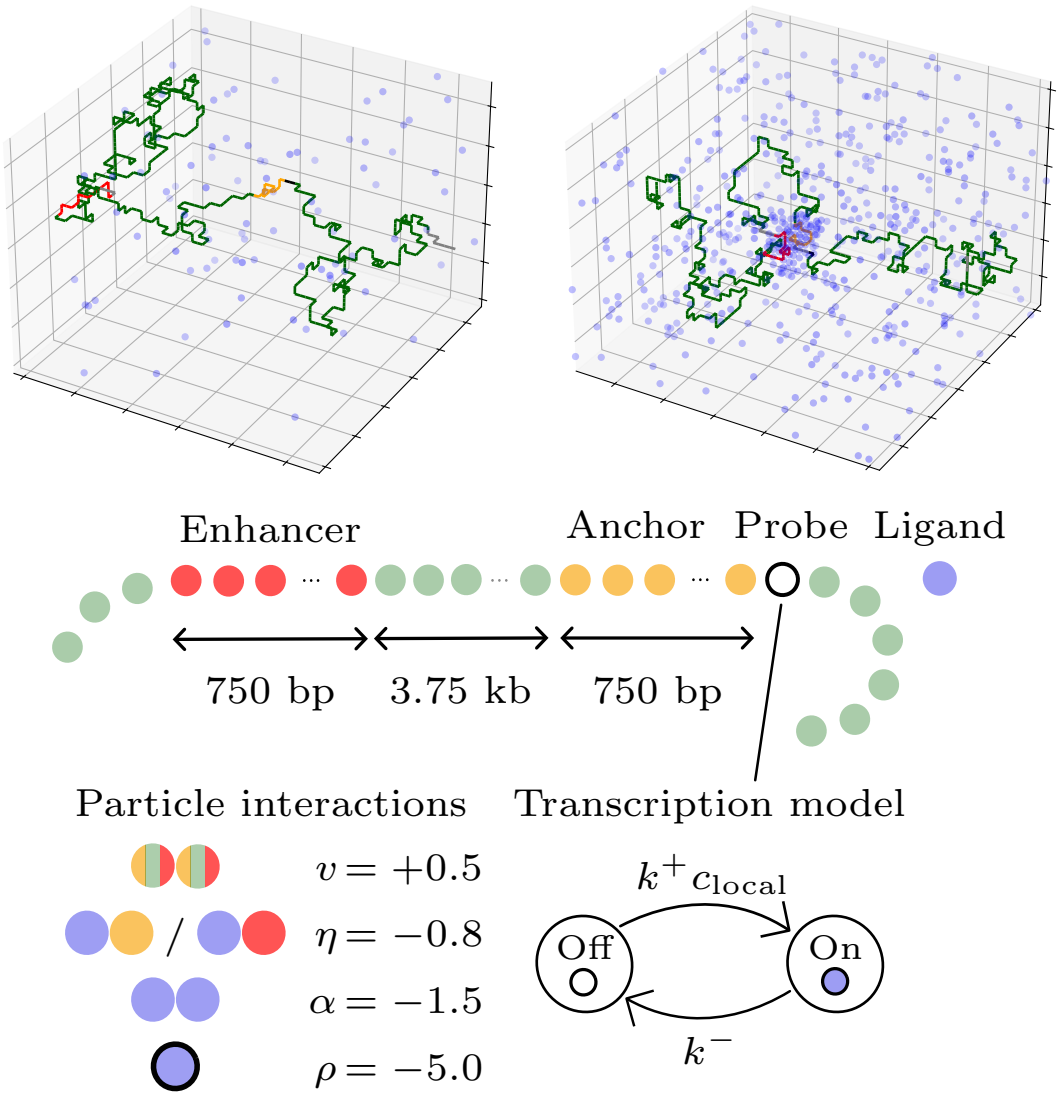
Rouches & Machta, *Biophysical Journal* 2025

Proteins condense on the surface of DNA

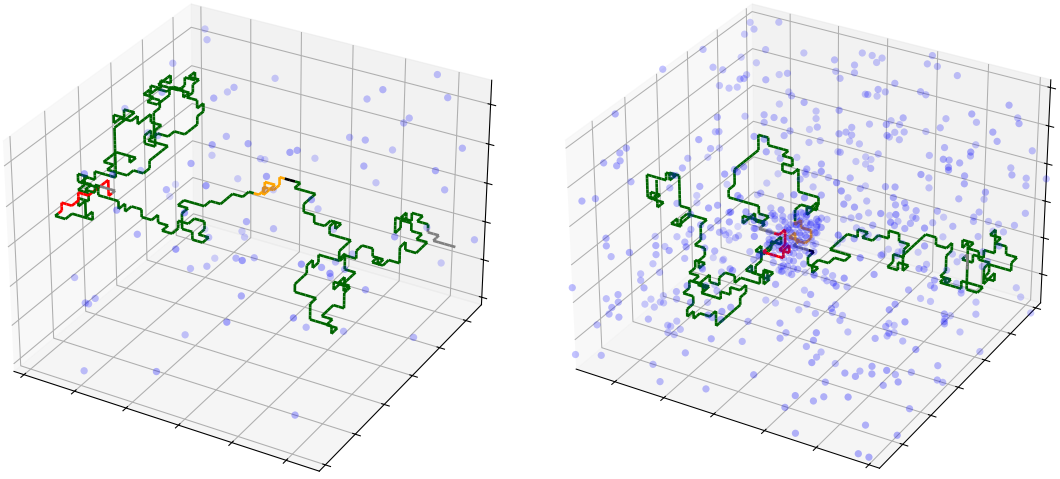


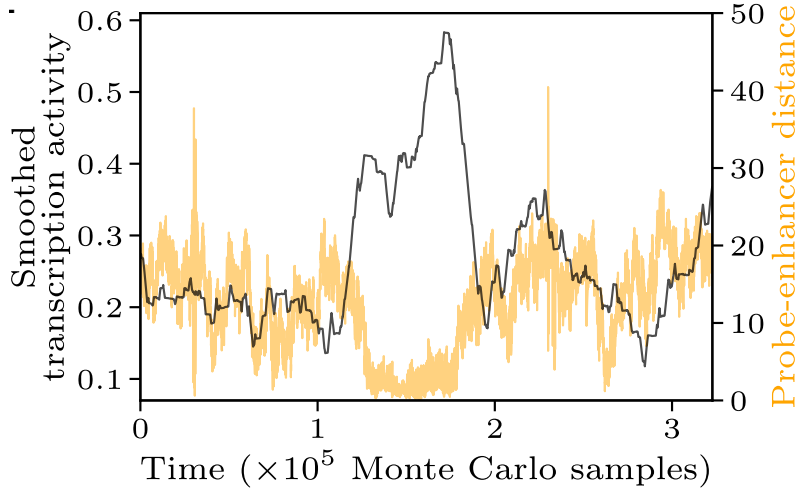
Influence on precision of gene expression?

Coupling condensation and polymer dynamics

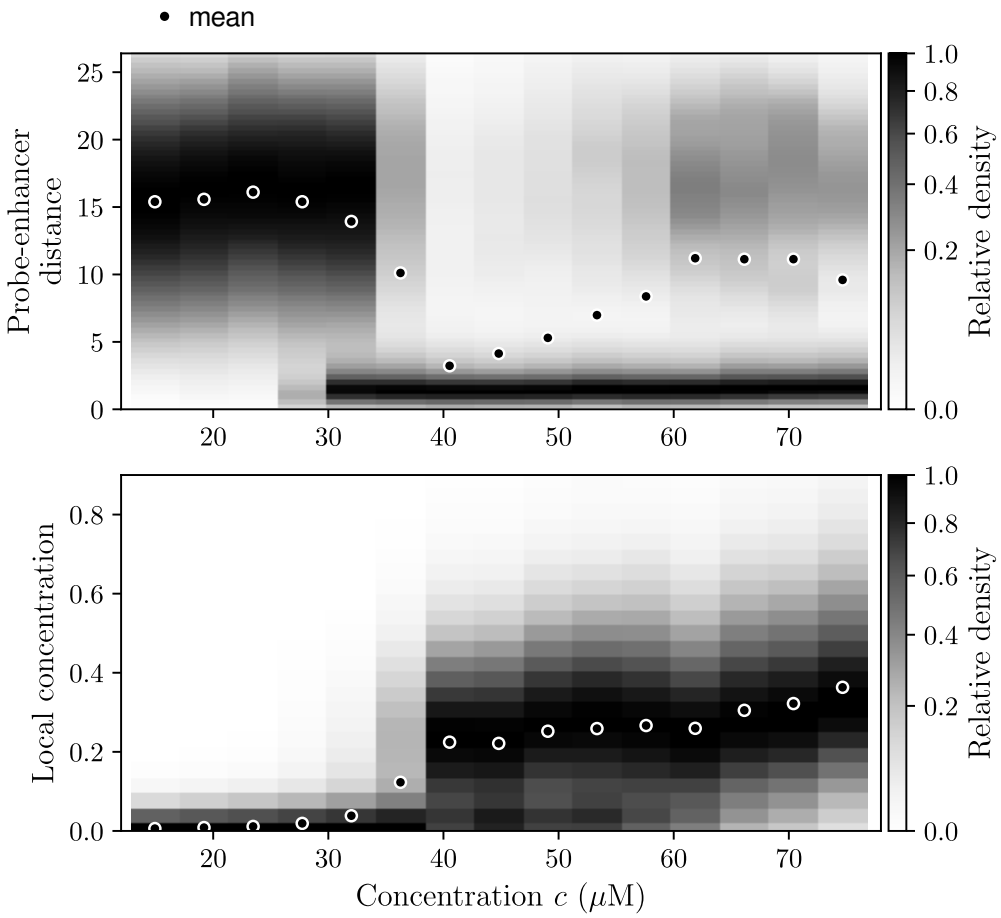


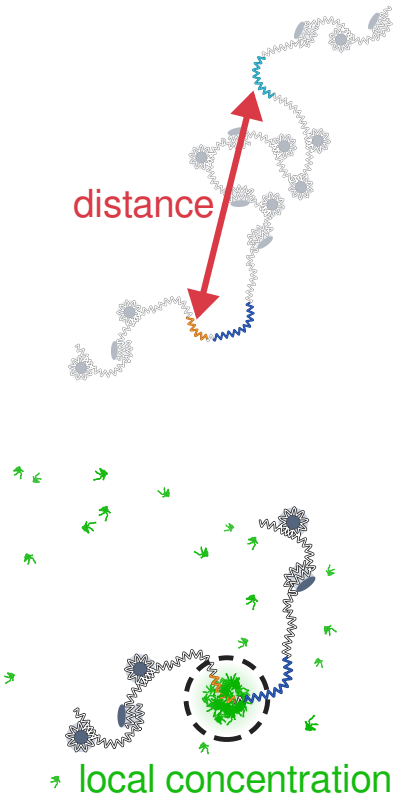
Coupling condensation and polymer dynamics



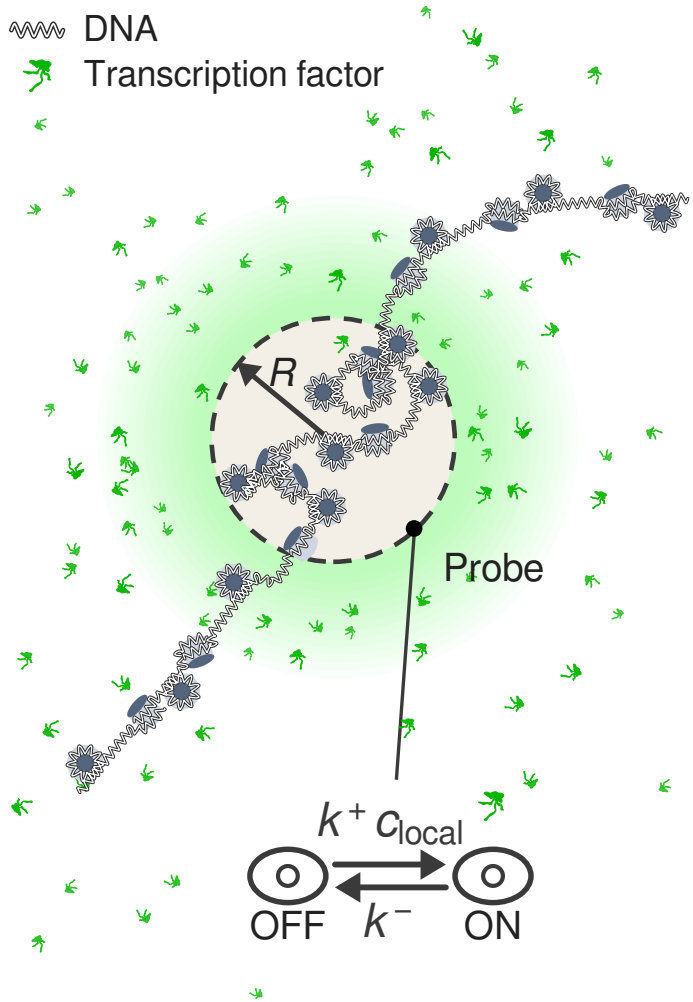


Polymer configuration depends sensitively on concentration





Continuous model of surface condensation



Flory-Huggins potential

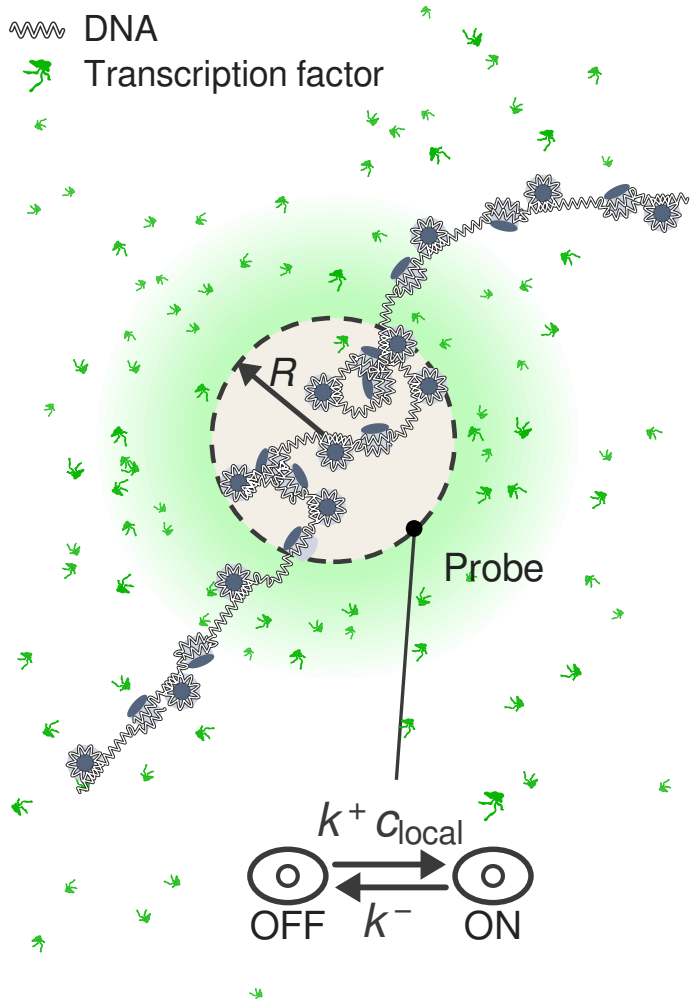
$$f(\phi) = \frac{\phi}{\nu}\ln\phi + (1-\phi)\ln(1-\phi) - \chi\phi^2 - \mu\phi$$

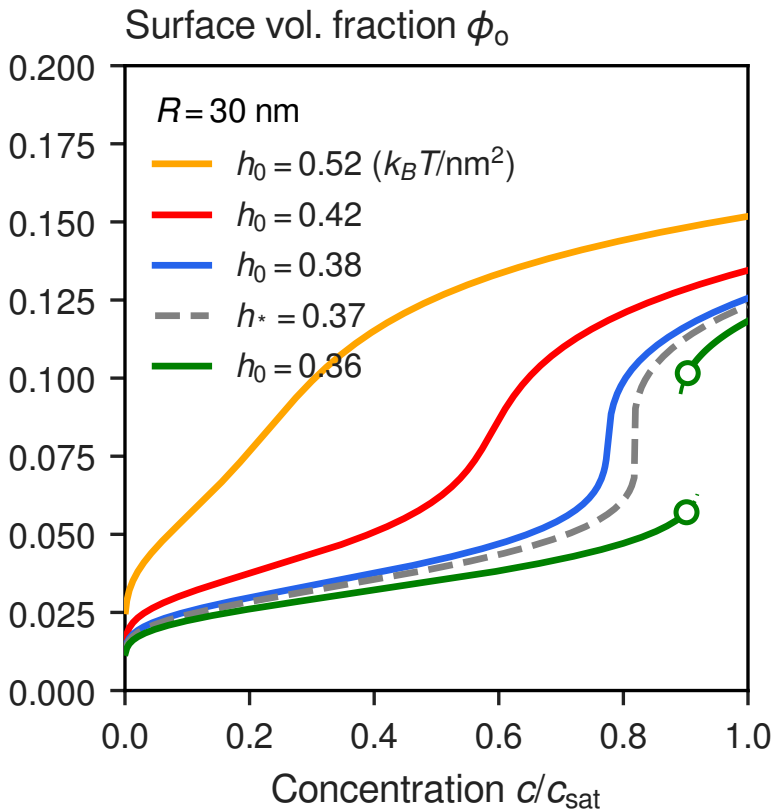
Cahn-Hilliard total energy

$$F = \int_B \left[\frac{\kappa}{2}(\nabla\phi)^2 + f(\phi)\right] d^3x - \int_S h_0\phi(\mathbf{x})dS$$

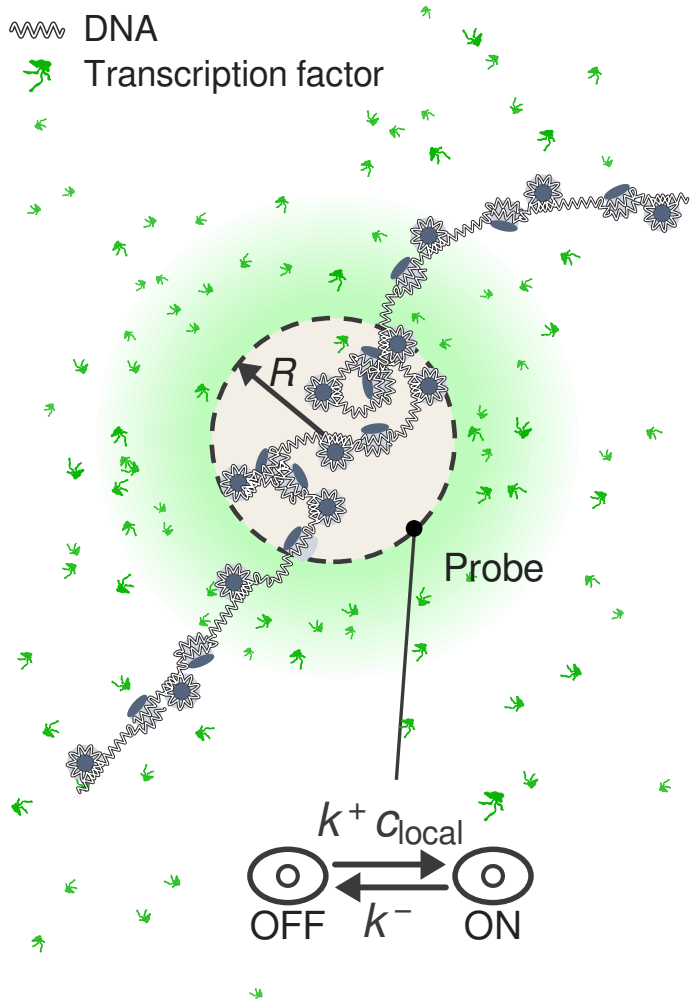
ϕ TF volume fraction
ν TF / solvent volume ratio
χ TF-TF attraction
μ chemical potential
h_0 surface attraction
κ Interface stiffness

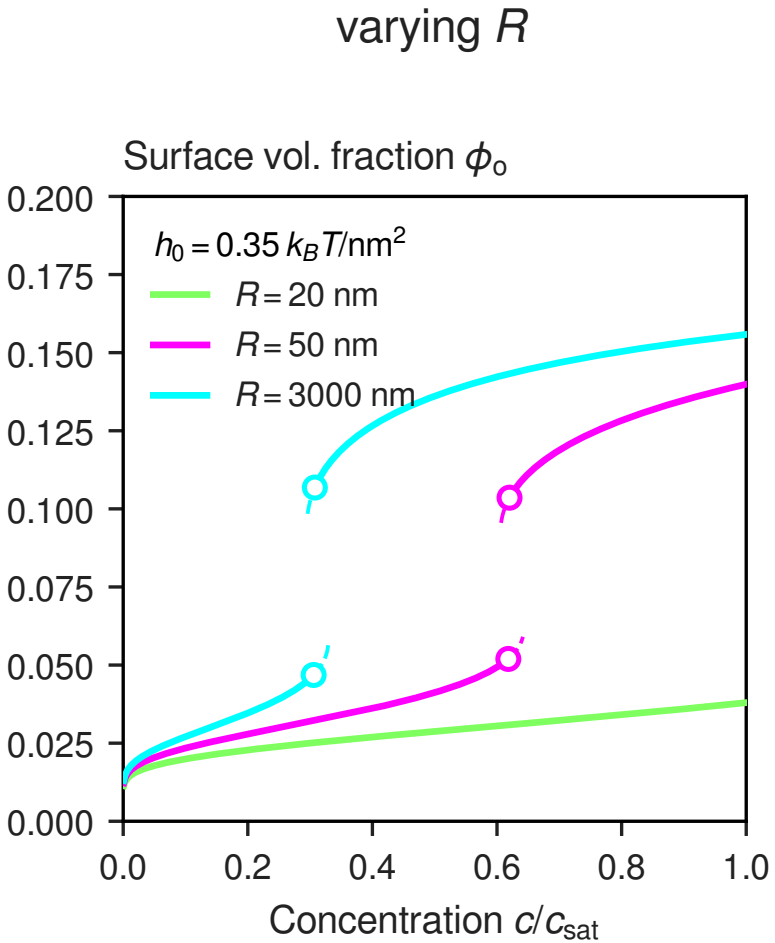
The prewetting transition: varying surface attraction



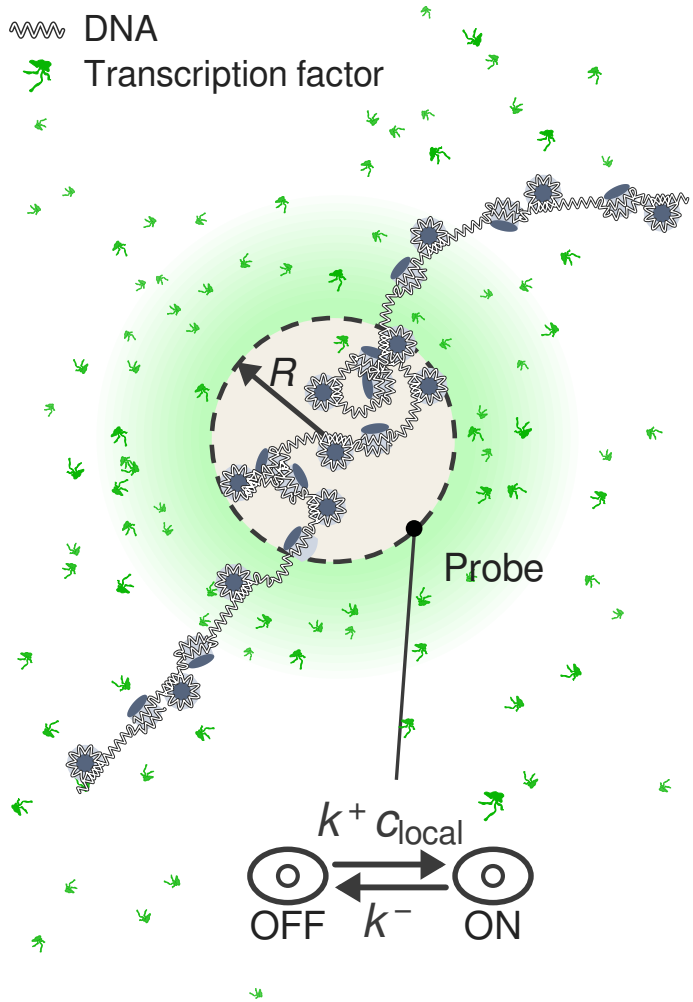


The prewetting transition: varying surface curvature





Quantifying sensitivity of local concentration



Free receptor

$$\dot{\mathcal{F}}_{\text{free}}(c) = \frac{1}{\tau_{\text{burst}}(c)}$$

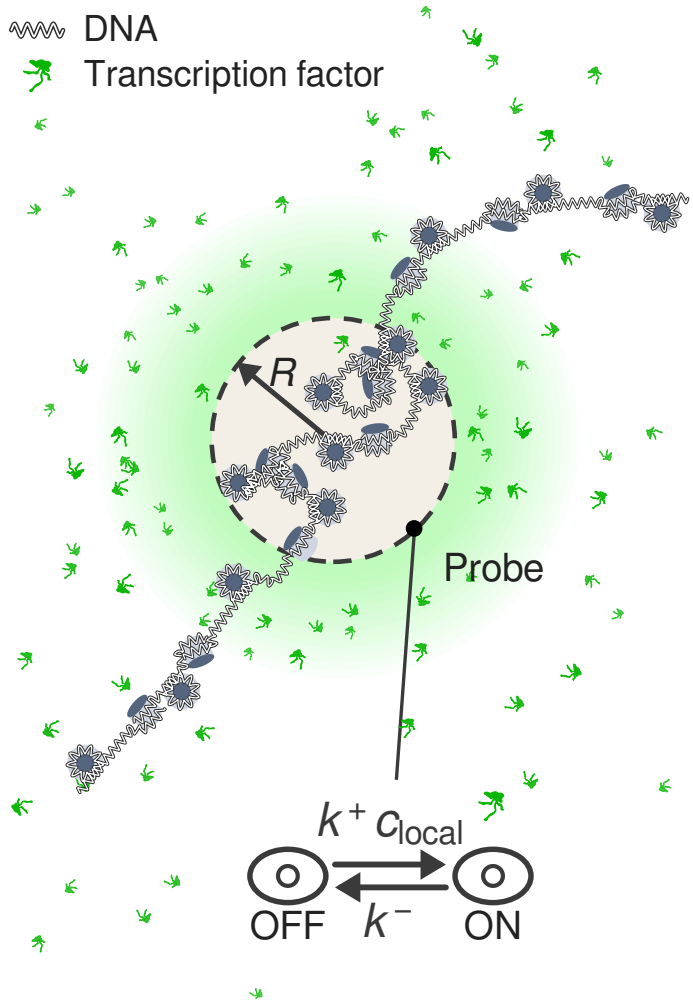
Receptor at the attractive surface

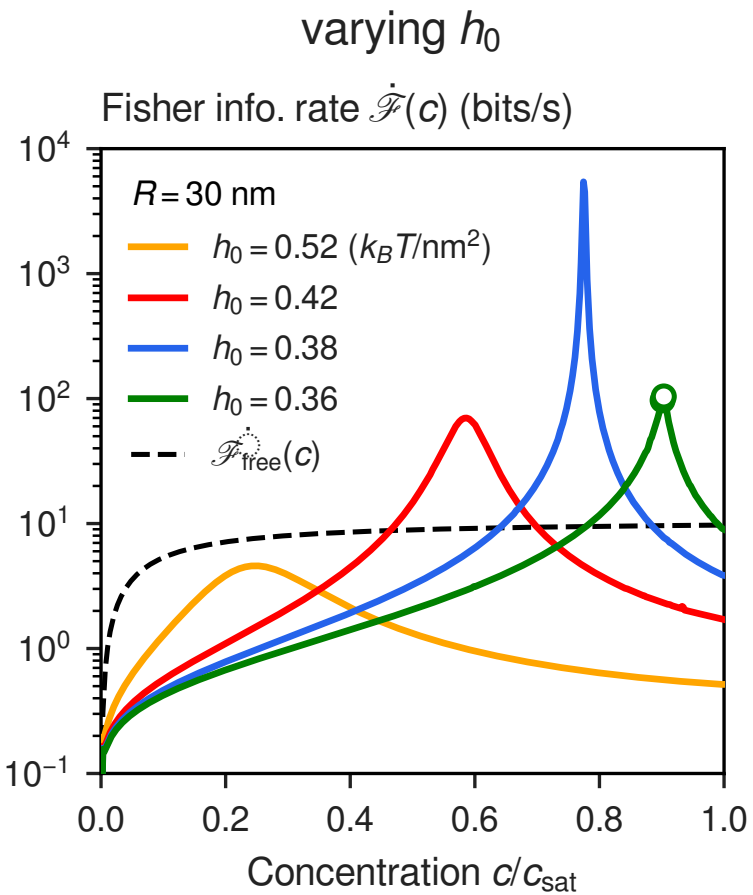
$$\dot{\mathcal{F}}(c) = \underbrace{\frac{1}{\tau_{\text{burst}}(c)}}_{\text{Probe}} \underbrace{\left(\frac{d\log c_o}{d\log c}\right)^2}_{\text{Surface}}$$

$\tau_{\text{burst}}(c)$ mean cycle time of the probe

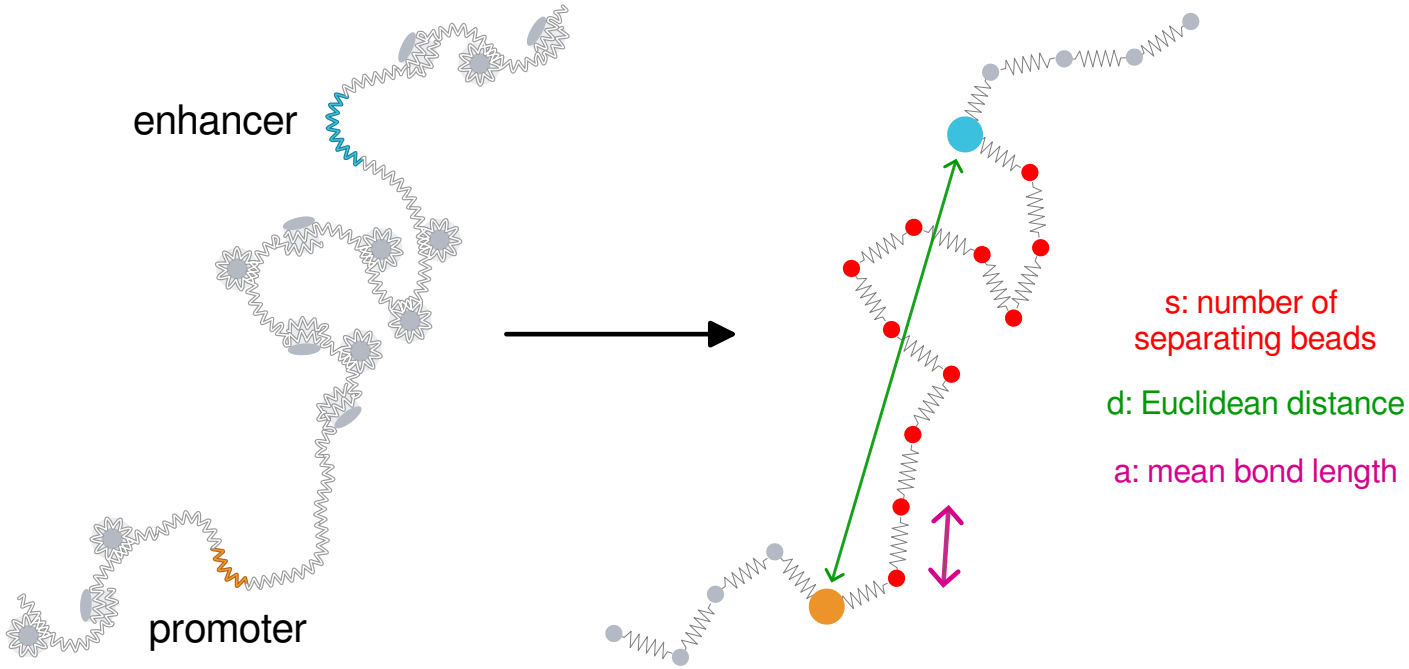
c_o TF volume fraction at the probe

Quantifying sensitivity of local concentration

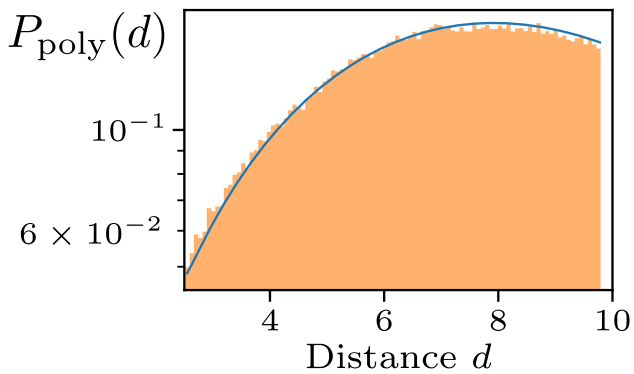




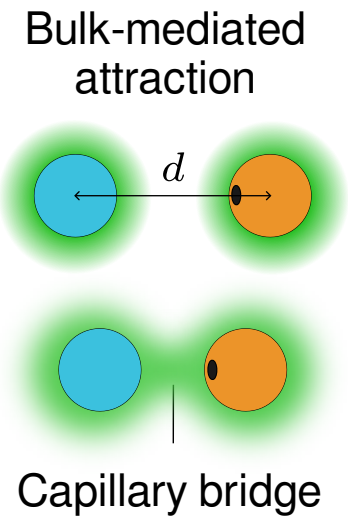
Rouse model of DNA



$$P_{\text{poly}}(d) \propto d^2 e^{-3d^2/2sa^2} \propto e^{-U_{\text{poly}}(d)/k_B T}$$



Promoter-enhancer attraction

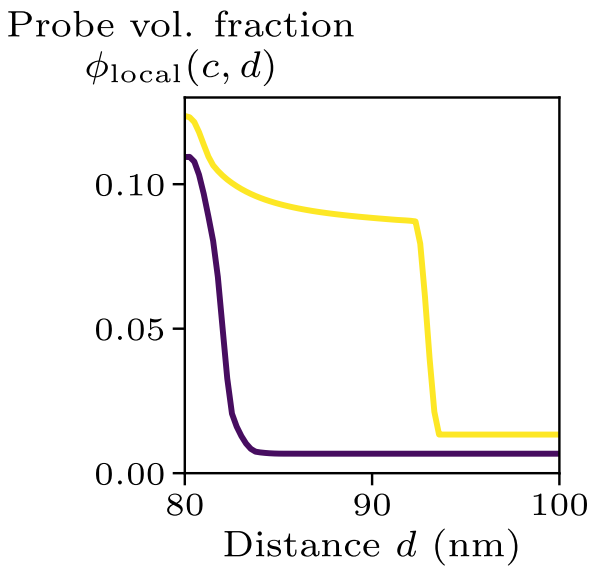


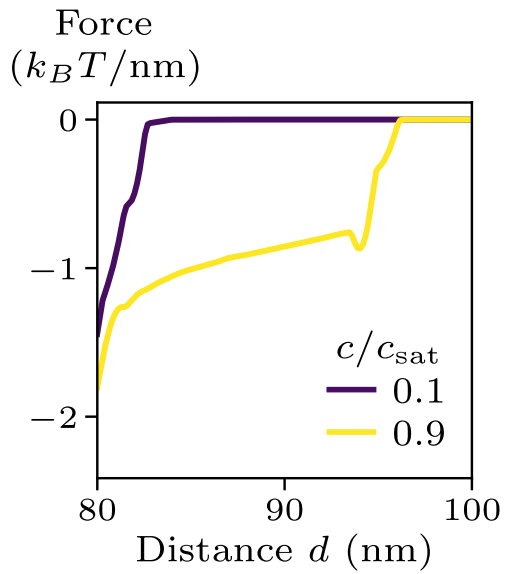
Cahn-Hilliard total energy

$$F = \int_B \left[\frac{\kappa}{2}(\nabla\phi)^2 + f(\phi)\right] d^3x - \int_{S_1} h_0\phi(\mathbf{x})dS - \int_{S_2} h_0\phi(\mathbf{x})dS$$

$$U_{\text{attr}}(d) = \min_{\phi(\mathbf{x})}\{F\}$$

$$\text{Force} = \partial_d U_{\text{attr}}(d)$$

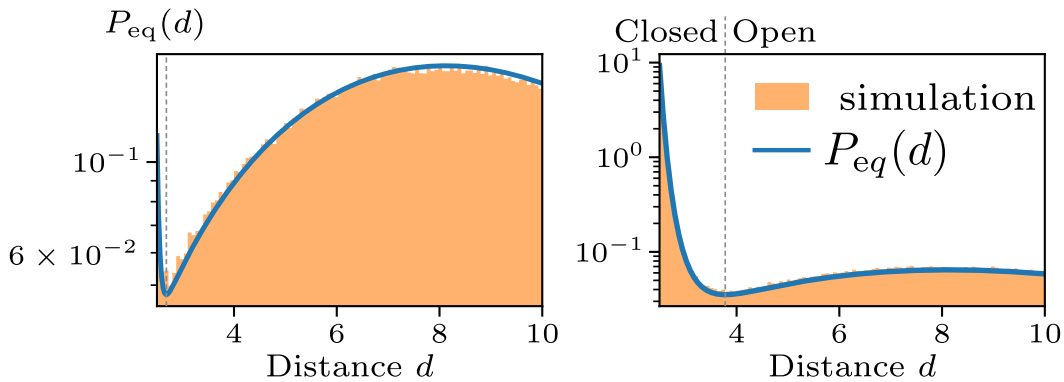




Polymer + capillary attraction

$$U_{\text{attr}}(d) + U_{\text{poly}}(d) = U_{\text{tot}}(d)$$

$$P_{eq}(d) \propto e^{-U_{tot}(d)/k_B T}$$

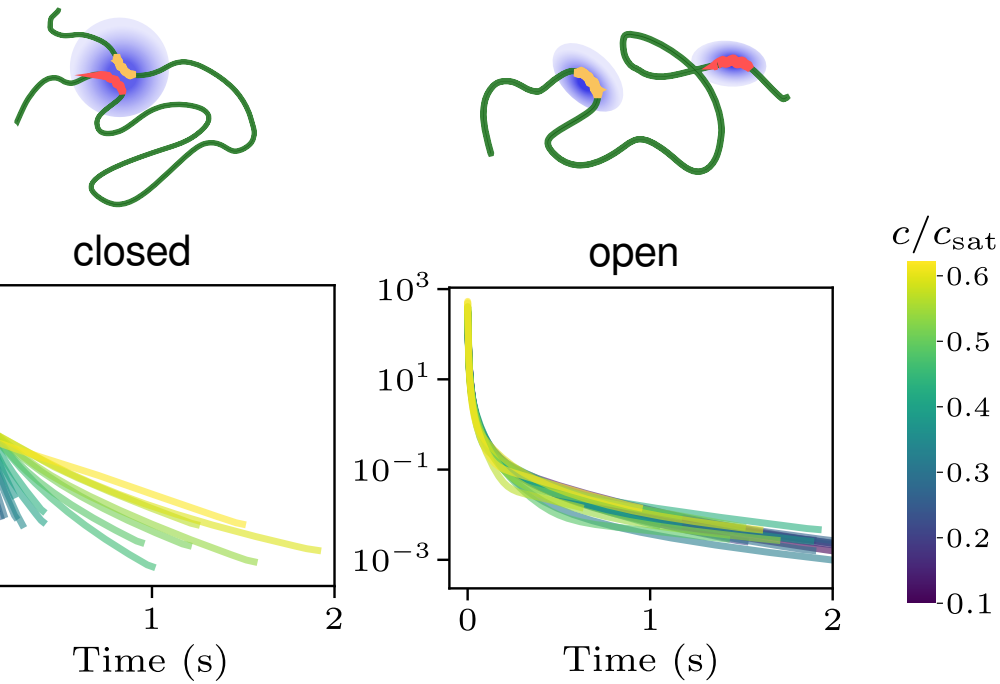


low concentration high concentration

Polymer dynamics

$$U_{\text{attr}}(d) + U_{\text{poly}}(d) = U_{\text{tot}}(d)$$

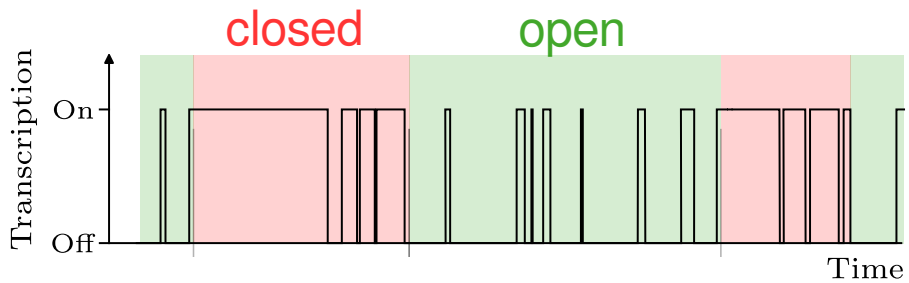
$$P_{eq}(d) \propto e^{-U_{tot}(d)/k_B T}$$



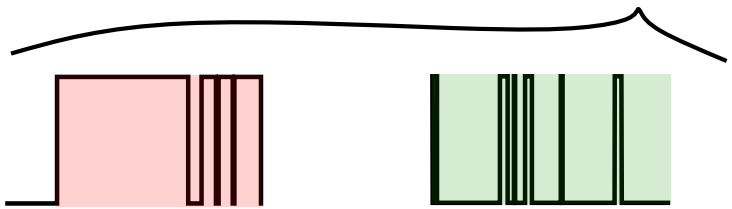
Sensitivity from polymer dynamics

$$\dot{\mathcal{F}}(c) = -\lim_{t\to\infty} \frac{1}{t} \left\langle \frac{\partial^2 \mathcal{L}(c)}{(\partial \ln c)^2} \right\rangle_c$$

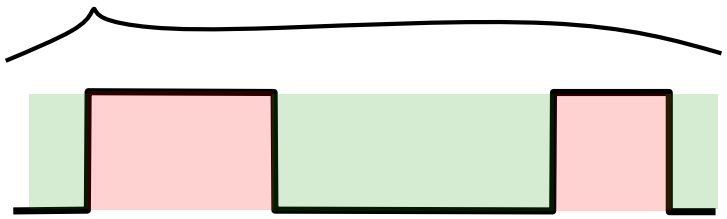
$$\mathcal{L}(c;t) = \ln P(\text{⎍⎍⎍⎍⎍}\,|c)$$

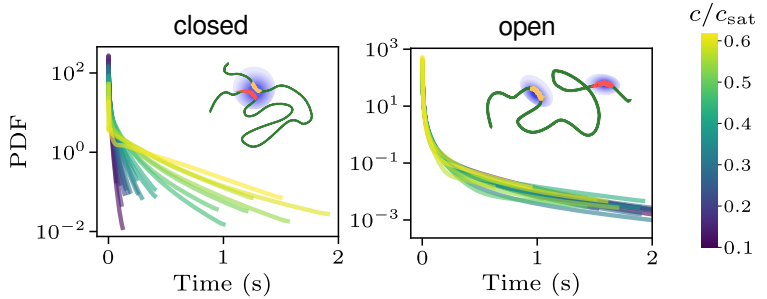


$$\dot{\mathcal{F}}(c) \approx \dot{\mathcal{F}}_{local}(c) + \dot{\mathcal{F}}_{DNA}(c)$$



$$\dot{\mathcal{F}}_o(c) = \underbrace{\frac{1}{\tau_{\text{burst}}(c)}}_{\text{Probe}} \underbrace{\left(\frac{d\log c_o}{d\log c}\right)^2}_{\text{Surface}}$$



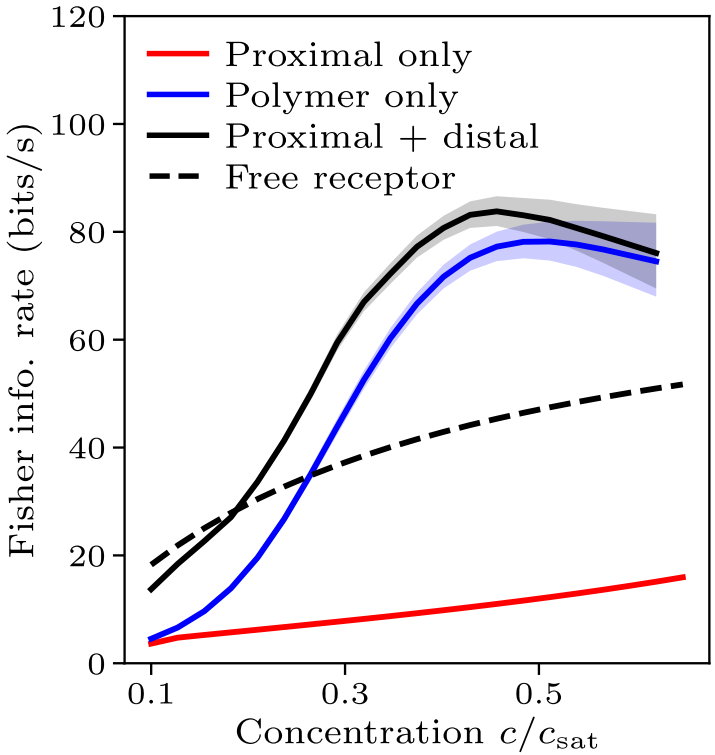


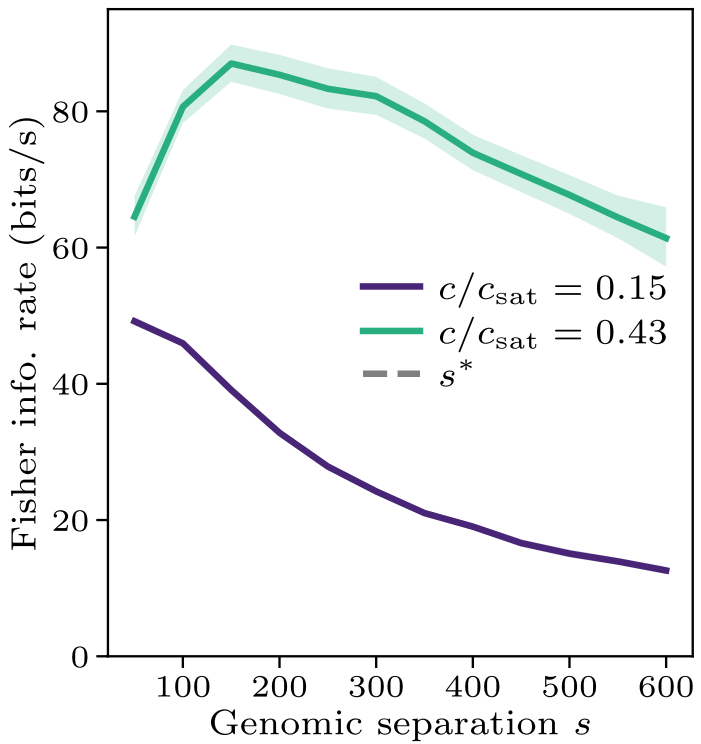
Total sensitivity

$$\dot{\mathcal{F}}(c) = -\lim_{t\to\infty} \frac{1}{t} \left\langle \frac{\partial^2 \mathcal{L}(c)}{(\partial \ln c)^2} \right\rangle_c$$

$$\mathcal{L}(c;t) = \ln P(\text{⎍⎍⎍⎍⎍} \,|c)$$

$$\dot{\mathcal{F}}(c) \approx \dot{\mathcal{F}}_{local}(c) + \dot{\mathcal{F}}_{\mathrm{DNA}}(c)$$





Takeaways

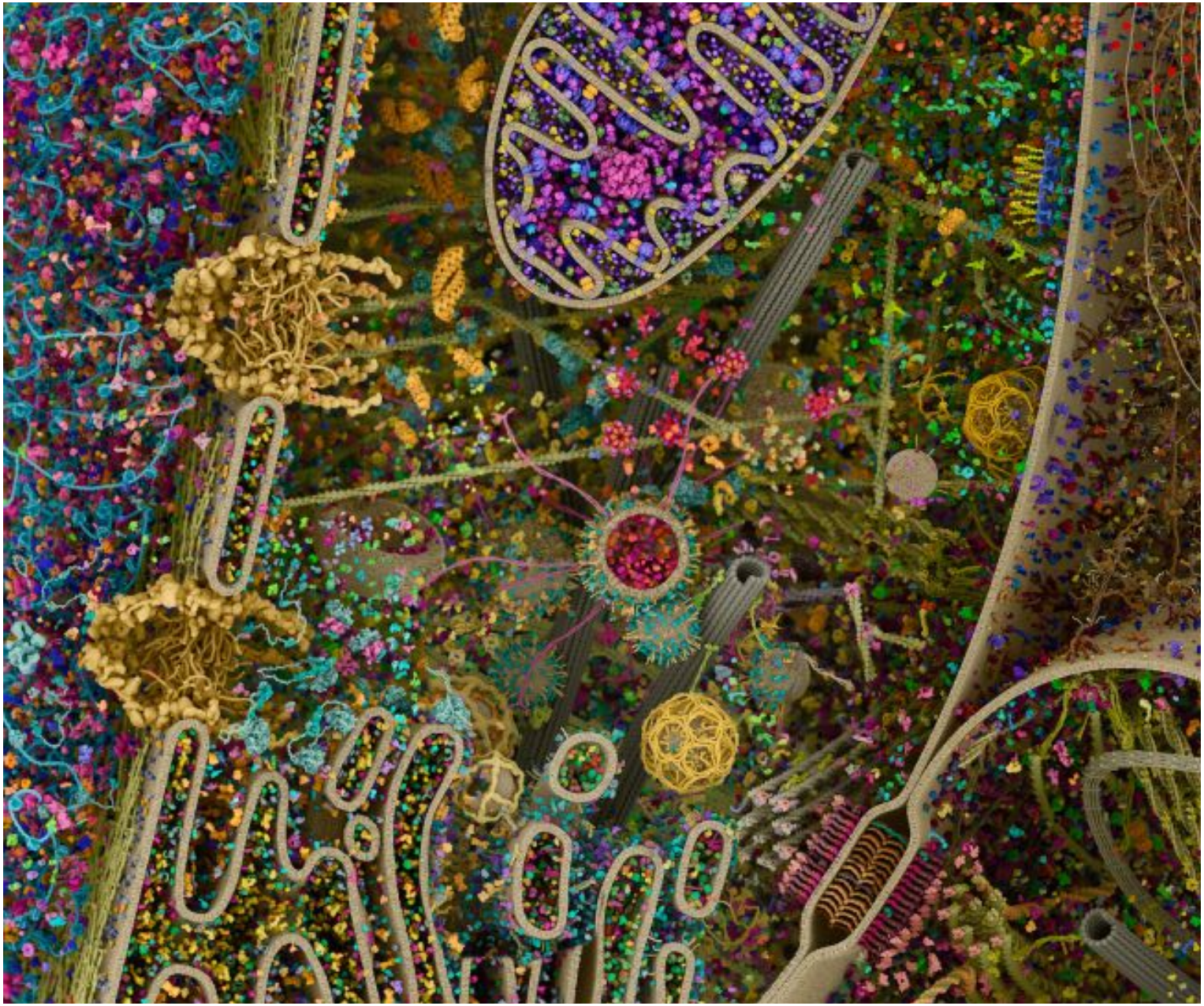
Surface condensation can boost concentration sensitivity

Polymer dynamics could contribute to gene expression sensitivity

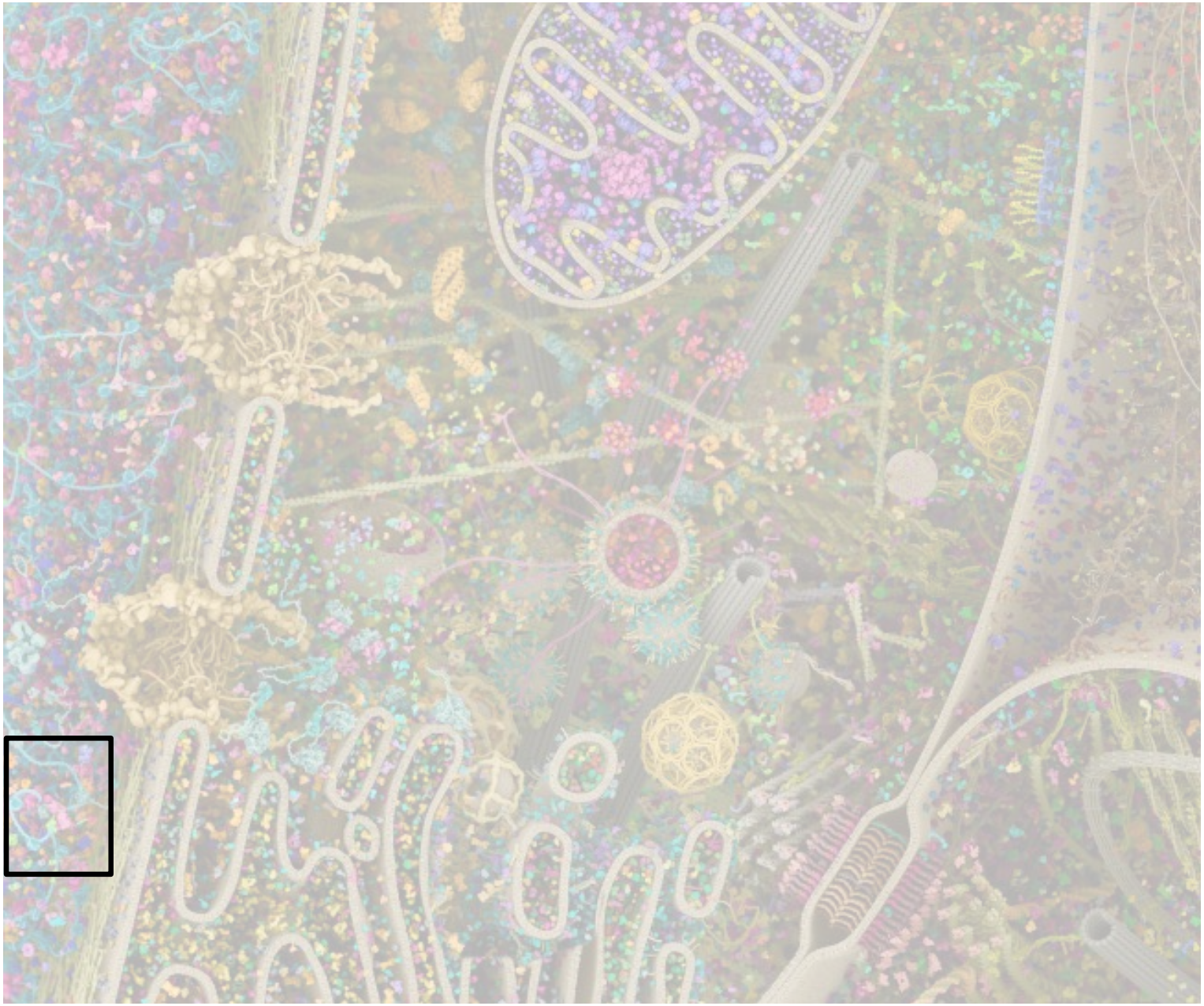
Limitations of the model:

- Rouse model unrealistic for DNA
- Loop extrusion, TADs
- Membrane binding behaves differently than bulk

Conclusion: Cells are complicated



Conclusion: Cells are complicated



Conclusion: Transcriptional regulation is very precise

Conclusion

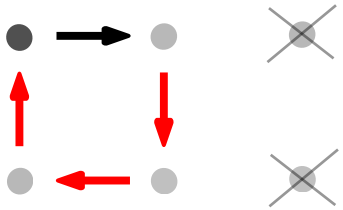
Q. What is the most concentration-sensitive promoter architecture?

sharp response?
cooperative?
proofreading?

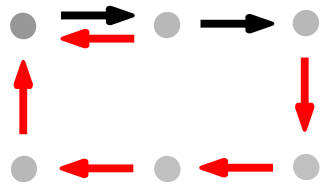
A. It depends on the limiting constraint

Cooperativity only optimal above critical concentration

"Keep binding sites free" (diffusion-limited)

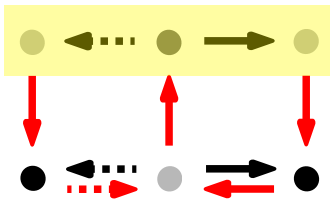


"Bind cooperatively" (activation-limited)

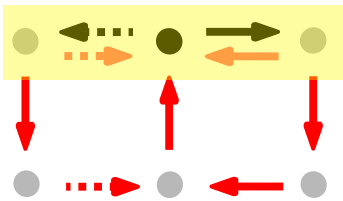


Proofreading only optimal above critical **decoy** concentration

"Permissive activation" (rare decoys)



"Proofreading" (abundant decoys)



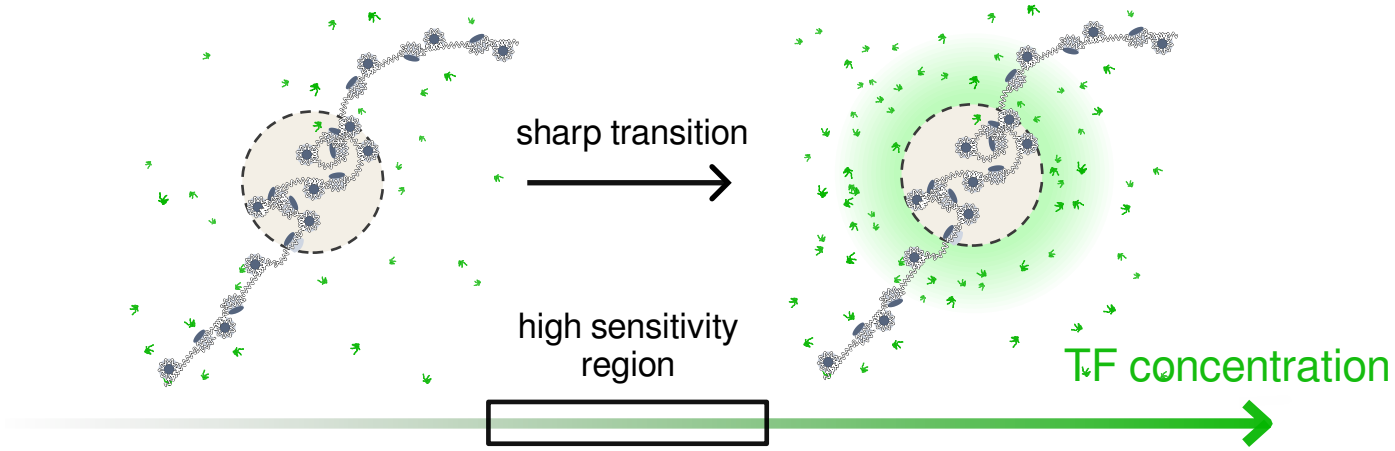
Solutions sets are generically **discontinuous**

Discussing optimality requires precise knowledge of underlying constraints

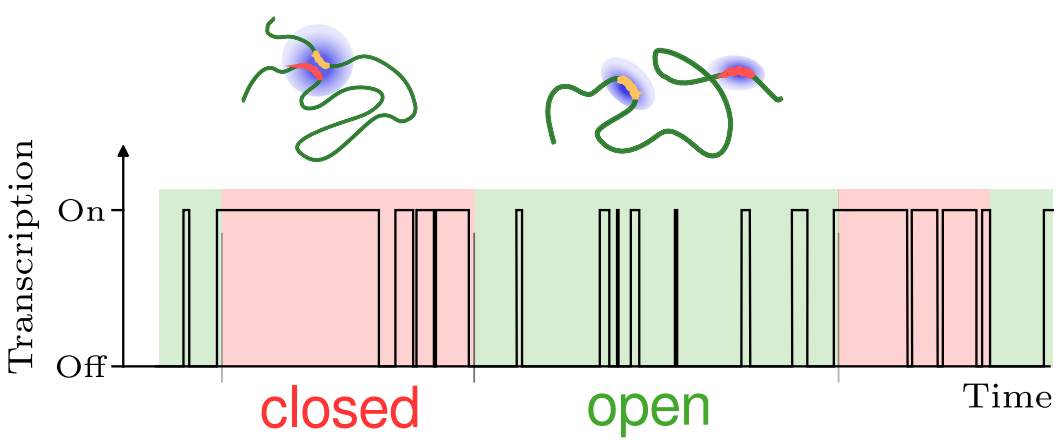
Conclusion

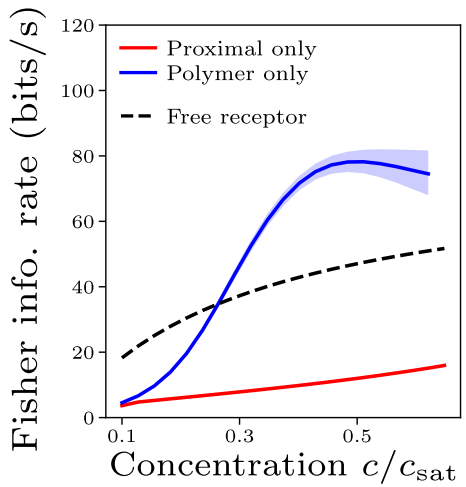
Q. How does a promoter's environment influence concentration sensitivity?

Surface condensation of transcription factors can amplify gene expression sensitivity



Polymer dynamics itself can encode a signal about concentration in transcription activity

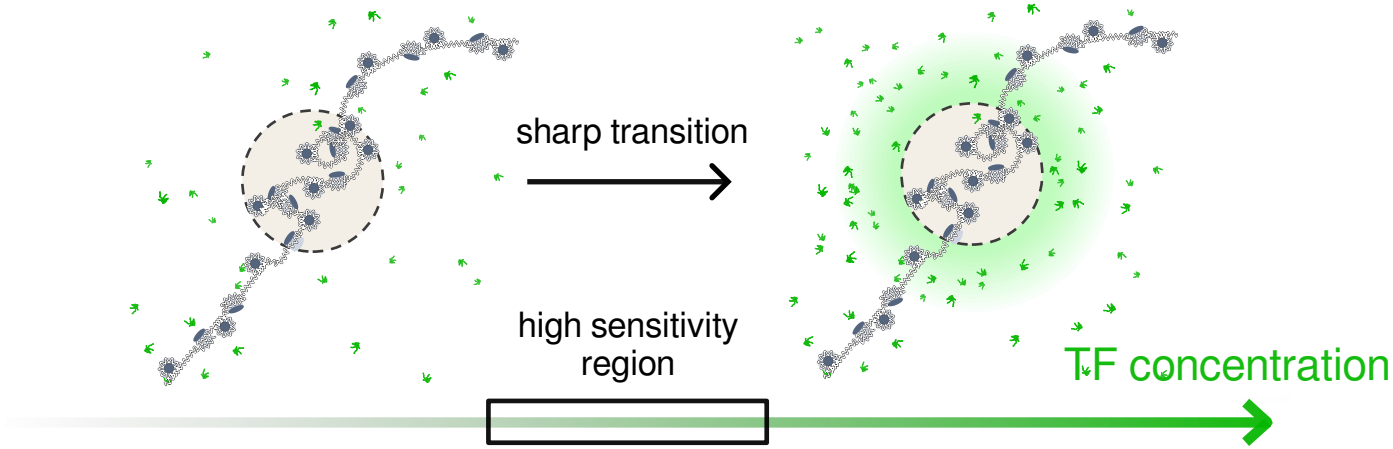




Thanks.

Q. How does a promoter's environment influence concentration sensitivity?

Surface condensation of transcription factors can amplify gene expression sensitivity



Polymer dynamics itself can encode a signal about concentration in transcription activity